ITCS448 Data Warehousing and Business Intelligence (2/2025)
Project Phase 2 Business Intelligence Implementation and Analytics

Project Title: Automotive
Business Case: Toyota
Group Number: 4
Group Members:

No.	Student ID	First name and Lastname	Role	Responsibilities
1.	6688072	Kemjira Nugboon	Data Modeler	Dimensional modeling, star schema design, data dictionary
2.	6688099	Nuttanan Reamprasert	Data Analyst	Oversee project progress, coordinate team tasks, analyze business insights
3.	6688129	Patsatraporn Thongdeesakul	BI Developer	Power BI dashboard development, visualization design
4.	6688168	Pornthita Sukprapaipat	Data Engineer	Data preparation, ETL process, Power Query cleaning and transformation
5.	6688210	Thanaporn Pinsakul	Project Manager	Manage project timeline, track deliverables, facilitate team communication, prepare presentation slides

Executive Summary

This project presents a Business Intelligence (BI) solution for analyzing Toyota dealership performance through the development of a data warehouse and interactive Power BI dashboards. The primary objective is to support data-driven decision-making by integrating and analyzing data across three key business domains: sales, service, and inventory.

A star schema data warehouse was designed to consolidate transactional data into structured fact and dimension tables, enabling efficient querying and analysis. The BI implementation includes multiple dashboards that visualize key performance indicators (KPIs) such as revenue trends, service activity, and inventory levels, allowing users to monitor business performance from different perspectives.

Key insights indicate that revenue fluctuates over time, with a decline in 2024 followed by recovery in 2025. Sales are highly concentrated among a small number of top-performing car models, suggesting dependency on specific products. Service operations provide a stable source of recurring revenue, while regional variations in service demand highlight potential inefficiencies. Additionally, inventory analysis reveals imbalances across dealerships, indicating opportunities for improved stock allocation.

The BI solution enhances decision-making by transforming raw data into actionable insights, improving operational visibility, and supporting strategic planning. Overall, this project demonstrates how data warehousing and BI tools can deliver meaningful business value and improve performance management within the automotive industry.

1. Introduction

1.1 Background of the Business Case

The automotive industry is highly competitive and continuously changing due to technological innovation, digital transformation, and evolving customer expectations. Companies in this industry must respond to global competition, sustainability trends, and increasing demand for data-driven decision-making. Toyota Motor Corporation is one of the world's leading automotive manufacturers, operating in more than 70 countries and producing a wide range of vehicles, including hybrid and electric cars. The company focuses on quality, innovation, and customer satisfaction to maintain its global competitiveness.

Toyota generates a large amount of data from various business activities such as vehicles sales, dealership operations, inventory management, customer information, and after sales services. However, this data is often stored in different systems across departments and regions, which makes it difficult to integrate and analyze effectively. As a result, decision-makers may face challenges in accessing timely and accurate insights. In today's digital era, organizations increasingly rely on business intelligence and data analytics to improve operational efficiency and support strategic planning. Therefore, implementing a data warehouse is essential to integrate data from multiple sources and provide a unified view of business performance. This project aims to design a data warehouse solution to support Toyota's business intelligence and decision-making processes.

1.2 Project Objectives

The main objective of this project is to design a data warehouse that supports business intelligence and improves decision-making for Toyota's automotive business. The objectives are divided into business and analytical perspectives as follows:

Business Objectives

- To improve visibility of sales performance across regions and dealerships.
- To support better inventory and supply chain management.
- To enhance customer relationship management and retention.
- To monitor after-sales service performance and warranty claims.
- To support strategic planning and business growth.

Analytical Objectives

- To identify key performance indicators (KPIs) related to sales, customers, and services.
- To analyze vehicle model performance and regional trends.
- To support customer behavior and repurchase analysis.
- To evaluate dealership performance and profitability.
- To enable data-driven decision making through dashboards and reports.

1.3 Project Scope and Assumptions

This project focuses on the analysis and design of a dimensional data warehouse for Toyota's automotive business. The scope includes business requirements analysis, source data analysis, dimensional modeling, and schema design.

However, the project does not include full system implementation or real-time data processing. The datasets used are obtained from public Kaggle sources and synthetic data created to simulate real-world scenarios due to data privacy limitations.

Key assumptions include:

- The datasets represent typical automotive business operations
- Synthetic data reflects realistic customer and dealership behavior
- The system is designed for analytical purposes rather than transaction processing
- Toyota has sufficient infrastructure to implement the solution in the future

2. Data Warehouse Analysis

2.1 Business Requirements Analysis

Stakeholders and Decision Makers

- Sales Managers: Monitor sales performance and track revenue targets.
- Dealership Managers: Evaluate dealership operations and overall profitability.
- Supply Chain Managers: Track inventory movement and manage stock shortages.
- Service Department Managers: Monitor service demand and track warranty-related issues.
- Executive Management: Focused on strategic planning and high-level performance evaluation.

Key Business Questions

- Which vehicle models generate the highest revenue?
- Which dealerships and salespersons generate the highest revenue?
- Which promotions and payment methods contribute most to total revenue
- What are the top service types generating the highest revenue?
- Which technicians perform most efficiently in terms of revenue and downtime?
- What is the trend and status distribution of service operations over time?
- Which models show signs of slow-moving inventory?
- Which suppliers demonstrate the shortest lead times?
- How is inventory distributed across dealerships?

2.2 Analytical Requirements

Key Performance Indicators (KPIs)

1. Sales Performance Focuses on revenue generation and sales efficiency.

- Average Order Value: The average total revenue per transaction to evaluate customer purchasing power
- Total Revenue: The sum of all sales transactions (sales_total) to monitor business growth.
- Sales Volume: Total quantity to track the number of units moved in a specific period.
- Total Orders: The total number of sales transactions recorded within a given period, used to track sales activity and demand volume.

2. Service & Maintenance Efficiency Focuses on workshop productivity and customer satisfaction.

- Average Service Price: The average charge per service transaction to monitor pricing consistency
- Total Service Revenue: Total income generated from maintenance and repairs (service_price).
- Average Downtime: Average downtime_days to track how long a customer's vehicle is out of service.
- Total Services: The total number of service transactions performed within a given period, used to monitor service workload and operational activity.

3. Inventory Management Focuses on stock health and availability.

- Average Supplier Lead Time: Average number of days from order to delivery per supplier to evaluate supply chain reliability
- Total Availability Qty: The ratio of available_qty vs. total stock to ensure sales readiness.
- Total Stock on hand: Comparing stock_in and stock_out to determine which models have the highest demand and optimize replenishment.

Measures (Facts)

Sales Measures:

- sales_quantity: The number of vehicle units sold per transaction.
- sales_price: The actual price at which the vehicle was sold after negotiations.
- discount_amount: The total value of discounts or promotional price reductions applied.
- sales_total: The total revenue generated per transaction (sales_price *sales_quantity).

Service Measures:

- service_price: Total service charge billed to the customer (service revenue).
- downtime_days: Number of days the vehicle remained in service.
- discount_amount: Total discount applied to the service charge.

Inventory Measures:

- stock_in: The number of units received into the warehouse from suppliers.
- stock_out: The number of units removed from inventory due to sales or transfers.
- stock_on_hand: The current balance of units available in physical inventory.
- reserved_qty: The number of vehicle units reserved by customers but not yet delivered.
- available_qty: The number of vehicle units available for sale, calculated from stock on hand minus reserved quantity.

Dimensions

- **Dim_Date**: Facilitates temporal analysis, allowing the CEO to monitor sales trends, service peaks, and inventory turnover on a daily, monthly, quarterly, and yearly basis.
- **Dim_Car**: Enables performance evaluation by vehicle characteristics such as model, color, transmission type, fuel type, engine size, and price segment (based on base_price).
- **Dim_Dealership**: Allows for regional sales comparison, branch performance benchmarking, and geographical analysis across different locations and store sizes.
- **Dim_Customer**: Supports customer segmentation analysis, demographic evaluation (occupation, income band), and tracking of long-term loyalty or repurchase behavior.
- **Dim_Salesperson**: Evaluates individual sales performance and commission tracking based on employee position, status, and hire date.
- **Dim_Promotion**: Measures the impact of different marketing campaigns and discount types on sales volume and revenue.

- **Dim_PaymentMethod**: Analyzes customer payment preferences, including bank-specific installments and interest rate impacts on sales.
- **Dim_Technician**: Monitors staff efficiency in the service department by tracking service quality, technician rank, and task completion time.
- **Dim_ServiceType**: Categorizes maintenance tasks (e.g., routine check, engine repair) and assigns priority levels to optimize workshop scheduling.
- **Dim_Supplier**: Enables supplier performance tracking, focusing on procurement volume, regional sourcing, and lead time reliability.
- **Dim_Location**: Provides granular details of specific storage areas or warehouse bins within the dealership to optimize internal logistics.

2.3 Source Data Analysis

In this project, the data used for analysis is primarily generated (synthetic data) based on structures and patterns derived from two main reference sources: the Toyota Dataset and the Car Sales Dataset from Kaggle. These reference datasets were used as guidelines to design realistic entities, attributes, and relationships in the operational (OLTP) database.

1. Generated Car Dataset (Based on Toyota Dataset)

- **Source Type:** Synthetic data (referenced from Toyota Dataset on Kaggle)
- **Description:** This dataset simulates vehicle specifications and pricing information. It is designed to represent Toyota car products and is used to construct the Car entity in the OLTP database.
- **Key Attributes (car):** The key attributes listed below reflect the original structure of the reference dataset. Additional attributes were generated and extended to align with the designed data warehouse schema.
    - car_id
    - model
    - year
    - price
    - transmission
    - fuel_type

The generated dataset supports vehicle-level analysis such as price comparison across models, year-based trend analysis, and segmentation by transmission and fuel type.

2. Generated Sales Dataset (Based on Car Sales Dataset)

- **Source Type:** Synthetic data (referenced from Car Sales Dataset on Kaggle)
- **Description:** This dataset simulates transactional sales records, including customer and dealership information. It is used to construct operational tables such as Sales Transaction, Dealership, and Customer in the OLTP schema.
- **Key Attributes (sales_transaction):** The key attributes listed below reflect the original structure of the reference dataset. Additional attributes were generated and extended to align with the designed data warehouse schema.
    - transaction_id
    - sale_date
    - car_id
    - quantity
    - total_amount

- **Key Attributes (customer):**
    - customer_id
    - gender
    - customer_segment

This dataset supports analysis of sales performance, dealership comparison, and customer behavior evaluation.

3. Additional Generated Datasets

To enhance the completeness of the business scenario, additional synthetic datasets were created to support all fact and dimension tables in the designed data warehouse schema. These include:

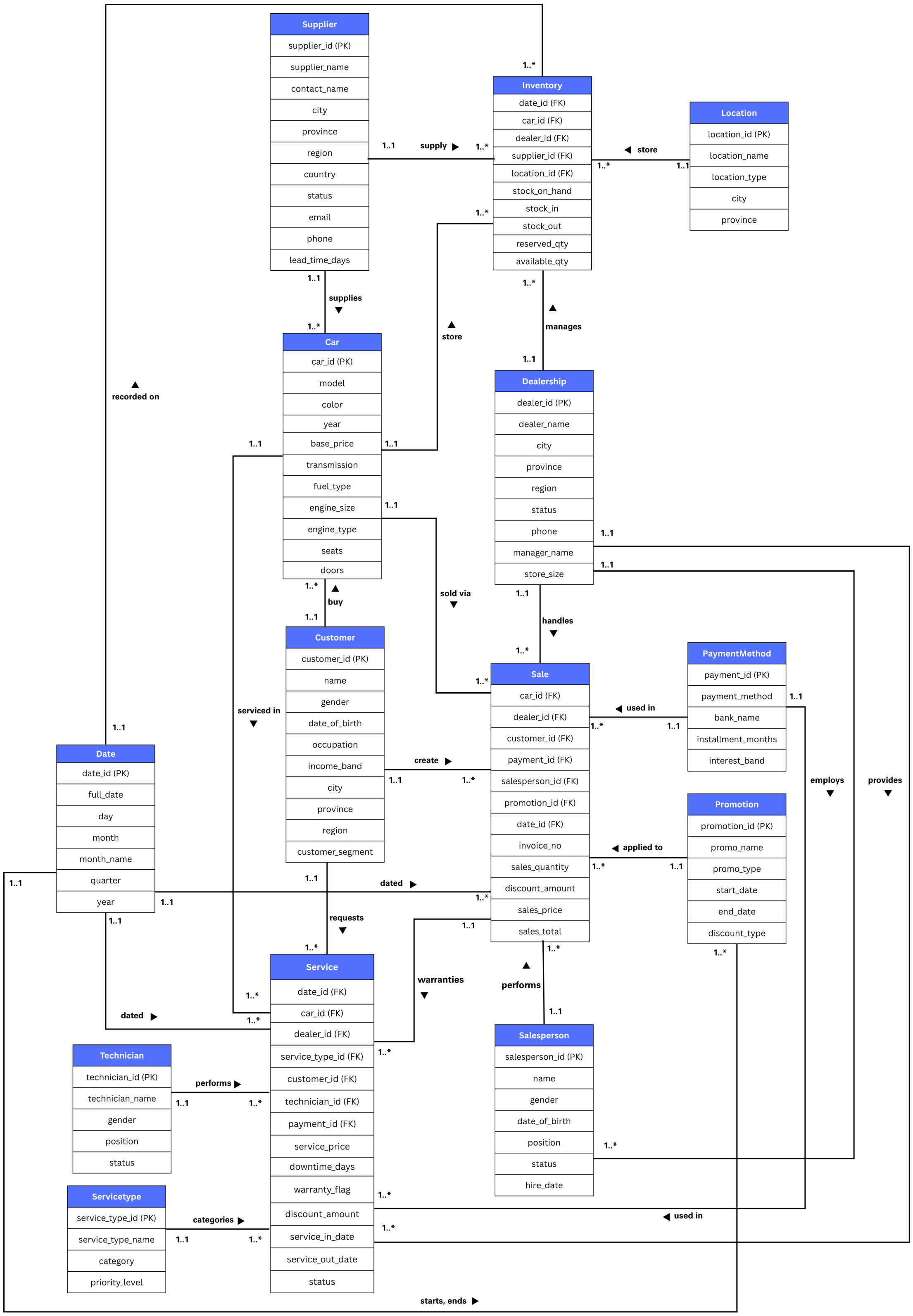
- **Dealership** (dealer information, location, store size, and operational status)
- **Customer** (demographics, income band, occupation, and customer segmentation)
- **Inventory** (stock levels, stock movement, and available quantity)
- **Service** (service transactions, downtime, warranty status, and service duration)
- **Salesperson** (employee information, position, and hire date)
- **Technician** (technician information, specialization, and employment status)
- **Supplier** (supplier details, regional sourcing, and lead time)
- **Location** (storage area and warehouse bin details within dealerships)
- **Promotion** (campaign details, promotion type, discount type, and validity period)
- **Payment Method** (payment type, bank information, installment terms, and interest band)
- **Service Type** (service category, service name, and priority level)

These datasets were generated due to privacy and accessibility limitations of real-world data. The structure and values were designed based on realistic automotive business assumptions to closely reflect actual dealership operations.

Data Characteristics

- Structured, relational data format suitable for OLTP systems
- Data volume covers a total of 14 tables and over 42,000 records across all entities, spanning the period from 2022 to 2025, simulating real-world transaction volumes. The largest tables include Sales (12,500 records), Service (19,000 records), and Inventory (6,311 records), while master data tables such as Car (322 records), Dealership (100 records), and Supplier (60 records) are smaller in size.
- Consistent keys and relationships across all entities to ensure referential integrity
- Includes both master data (Car, Customer, Dealership, Supplier, Technician, Salesperson) and transactional data (Sales, Service, Inventory)

ER-Diagram (OLTP Overview)



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3. Data Warehouse Design

3.1 Dimensional Modeling

Grain Definition

- **Sales Fact:** One car sale transaction for one customer, on one date, at one dealership, handled by one salesperson, with one promotion and one payment method.
- **Service Fact:** One service transaction for one car, performed for one customer, on one date, at one dealership, handled by one technician, for one service type, with one payment method.
- **Inventory Fact**: One inventory record for one car, at one dealership, in one specific location, supplied by one supplier, on one date.

Fact Tables

Our model includes three primary fact tables to address key business questions:

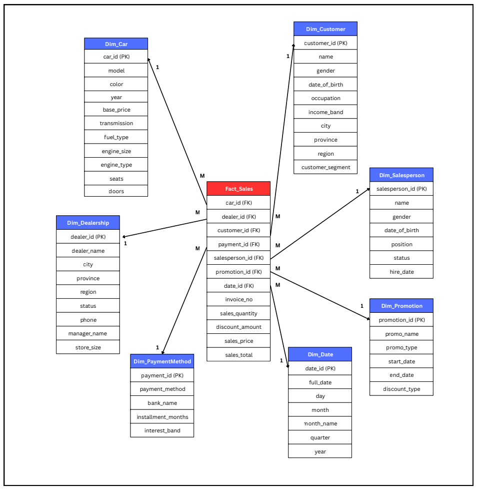
- **Fact_Sales:** invoice_no, sales_quantity, discount_amount, sales_price, discount_amount ,sales_total
- **Fact_Service:** service_price, downtime_days, warranty_flag, discount_amount, service_in_date, service_out_date, status
- **Fact_Inventory:** stock_on_hand, stock_in, stock_out, reserved_qty, available_qty

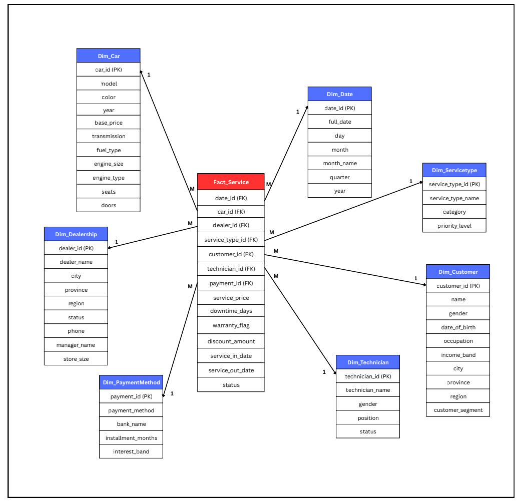
Dimension Tables

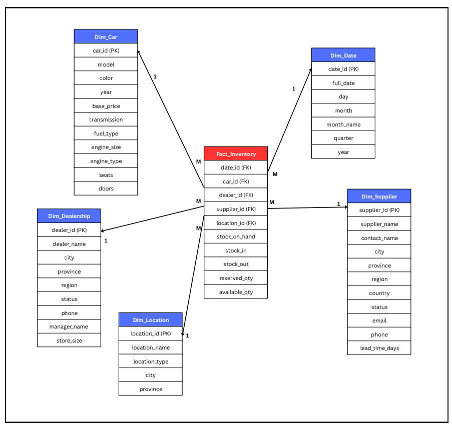
To provide contextual information for analysis, we utilize eleven main dimensions:

- **Dim_Date:** date_id, full_date, day, month, month_name, quarter, year
- **Dim_Car:** car_id, model, color, year, base_price, transmission, fuel_type, engine_size, engine_type, seats, doors
- **Dim_Dealership:** dealer_id, dealer_name, city, province, region, status, phone, manager_name, store_size
- **Dim_Customer:** customer_id, name, gender, date_of_birth, occupation, income_band, city, province, region, customer_segment
- **Dim_Salesperson:** salesperson_id, name, gender, date_of_birth, position, status, hire_date
- **Dim_PaymentMethod:** payment_id, payment_method, bank_name, installment_months, interest_band
- **Dim_Promotion:** promotion_id, promo_name, promo_type, start_date, end_date, discount_type
- **Dim_Technician:** technician_id, technician_name, gender, position, status
- **Dim_ServiceType:** service_type_id, service_type_name, category, priority_level
- **Dim_Supplier:** supplier_id, supplier_name, contact_name, city, province, region, country, status,email,phone, lead_time_days
- **Dim_Location:** location_id, location_name, location_type, city, province

3.2 Data Warehouse Schema



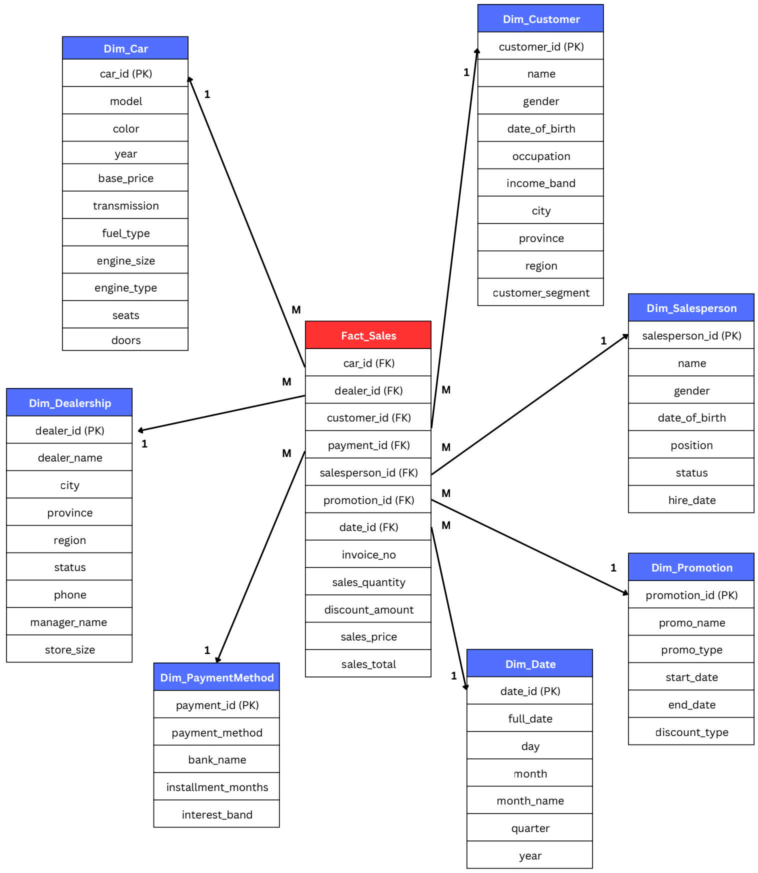




[Link for full star schema]

This overall star schema represents an integrated automotive dealership (TOYOTA) data warehouse consisting of three fact tables: Fact_Sales, Fact_Service, and Fact_Inventory. It connects core business processes vehicle sales, after-sales service, and inventory management through shared dimensions such as Car, Date, and Dealership. The model enables comprehensive analysis of revenue, service performance, stock levels, customer behavior, operational efficiency, and supplier management, supporting strategic decision-making across the entire dealership operation.

1) Fact_Sales Star Schema



This Star Schema is designed to analyze Toyota's sales performance by capturing granular transactional data, including units sold, prices, and discounts. By integrating dimensions like Dim_Date, Dim_Car, and Dim_Customer, the system provides a view of regional revenue trends and vehicle model performance. This design transforms raw sales records into actionable insights to support strategic decision-making and profitability tracking.

Grain:
One car sale transaction for one customer, on one date, at one dealership, handled by one salesperson, with one promotion and one payment method.

Measures:

- sales_quantity
- discount_amount
- sales_price (unit price)
- sales_total (final revenue after discount)

Degenerate Dimension:

- invoice_no (transaction identifier stored directly in the fact table)

Connected Dimensions and Analytical Purpose:

- **Dim_Date**
  Supports sales trend analysis over time (daily, monthly, quarterly, yearly).

- **Dim_Car**
  Enables performance comparison by vehicle model, transmission type, fuel type, and manufacturing year.

- **Dim_Customer**
  Supports customer segmentation, demographic analysis, and repurchase behavior tracking.

- **Dim_Dealership**
  Allows regional sales comparison and dealership performance evaluation.

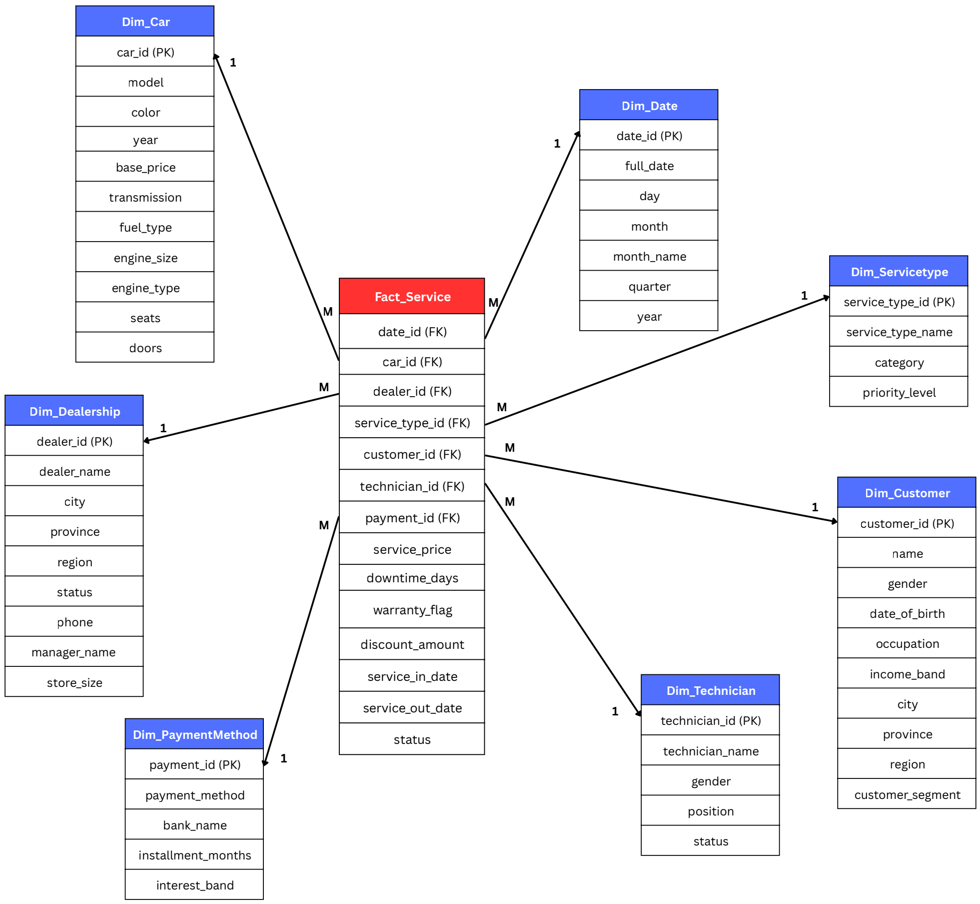
- **Dim_Salesperson**
  Enables salesperson performance analysis, commission tracking, and productivity comparison.

- **Dim_Promotion**
  Supports evaluation of promotion effectiveness and discount impact on sales performance.

- **Dim_PaymentMethod**
  Allows analysis of customer payment preferences (cash, financing, leasing, etc.) and revenue distribution by payment type.

The overall business purpose of this schema is to analyze vehicle sales performance across products, customers, locations, and time to support revenue growth, profitability improvement, and strategic decision-making.

2) Fact_Service Star Schema



The Fact_Service Star Schema monitors the operational efficiency and financial health of the after-sales department by tracking service revenue, labor duration, and downtime. By analyzing these metrics against Dim_Technician and Dim_ServiceType, management can evaluate technician productivity and service category demand. This framework enables Toyota to optimize resource allocation and enhance customer retention through improved service quality.

Grain:
One service transaction for one car, performed for one customer, on one date, at one dealership, handled by one technician, for one service type, with one payment method.

Measures:

- service_price (unit price)
- downtime_days
- discount_amount

Degenerate Dimension:

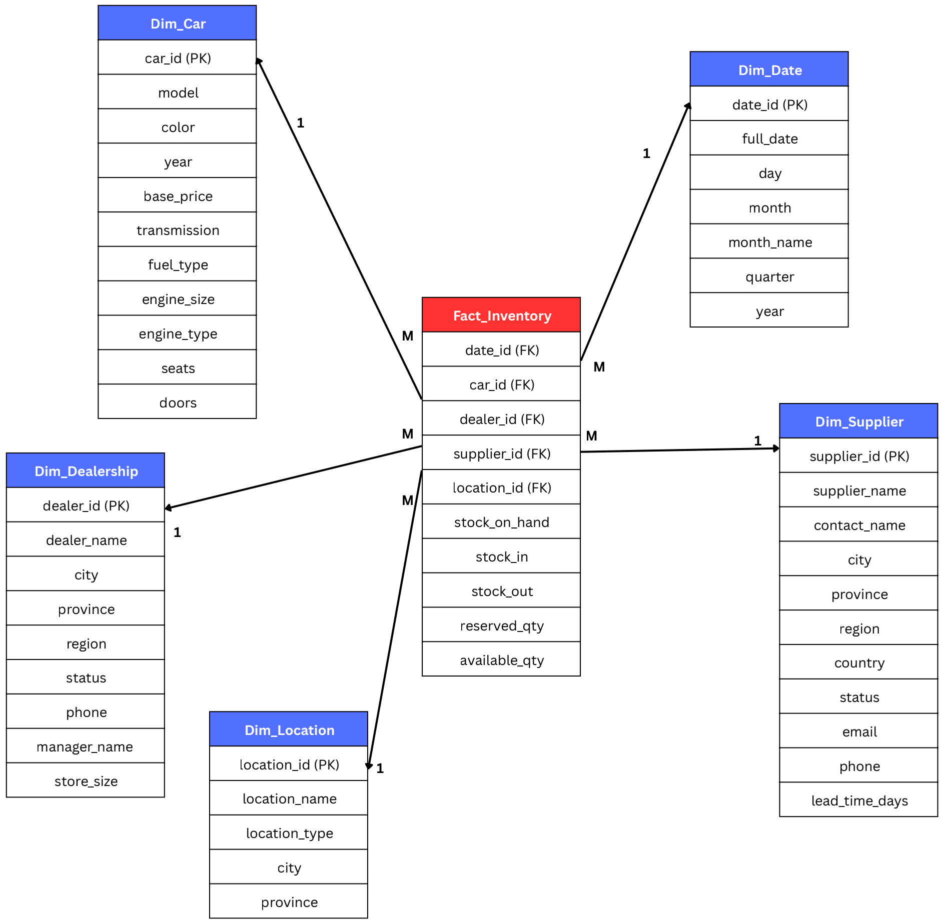
- warranty_flag (Service status indicator used to distinguish between warranty and paid services)
- service_in_date (Service start date used to track when a vehicle entered the service process)
- service_out_date (Service completion date used alongside service_in_date to calculate actual service duration and support downtime analysis)

Connected Dimensions and Analytical Purpose:

- **Dim_Date**
  Supports service trend analysis over time (daily workload, monthly trends, seasonal patterns, yearly growth).
- **Dim_Car**
  Enables analysis of service frequency by car model, fuel type, transmission type, and manufacturing year.
- **Dim_Customer**
  Supports customer retention analysis, service frequency tracking, and after-sales behavior insights.
- **Dim_Dealership**
  Allows comparison of service revenue and operational performance across dealership locations.
- **Dim_Technician**
  Enables technician productivity analysis, performance comparison, and labor efficiency evaluation.
- **Dim_ServiceType**
  Supports analysis of service category performance (maintenance, repair, warranty, inspection, etc.).
- **Dim_PaymentMethod**
  Allows analysis of customer payment preferences and revenue distribution by payment type.

This schema enables analysis of service revenue, technician performance, service trends over time, dealership efficiency, customer retention, and payment behavior.

3) Fact_Inventory Star Schema



This Star Schema serves as a periodic snapshot for tracking stock levels and optimizing supply chain logistics across Toyota's dealership network. The central Fact_Inventory table monitors stock movement including stock-in, stock-out, and available quantities to maintain high visibility of asset value. This approach allows for the identification of slow-moving inventory and the evaluation of supplier reliability to minimize carrying costs.

Grain:
One inventory record for one car, at one dealership, in one specific location, supplied by one supplier, on one date.

Measures:

- stock_on_hand
- stock_in
- stock_out
- reserved_qty
- available_qty

Connected Dimensions and Analytical Purpose:

- **Dim_Date**
  Supports inventory trend analysis over time and stock aging evaluation.

- **Dim_Car**
  Enables analysis of stock levels by car model, type, and year.

- **Dim_Dealership**
  Allows comparison of inventory levels across dealership branches.

- **Dim_Location**
  Supports warehouse or storage-level inventory tracking.

- **Dim_Supplier**
  Enables supplier performance evaluation and supply chain analysis.

The scheme is to track vehicle stock levels across dealerships over time to optimize inventory control and cost management.

3.3 Data Dictionary

Fact Tables

1) Fact_Sales

Table	Attribute	Type	Format	Key	Reference	Description
Fact_Sales	date_id	INT	11	FK	Dim_Date (date_id)	Links to Dim_Date
Fact_Sales	car_id	INT	11	FK	Dim_Car (car_id)	Links to Dim_Car
Fact_Sales	dealer_id	INT	11	FK	Dim_Dealership (dealer_id)	Links to Dim_Dealership
Fact_Sales	customer_id	INT	11	FK	Dim_Customer (customer_id)	Links to Dim_Customer
Fact_Sales	payment_id	INT	11	FK	Dim_PaymentMethod (payment_id)	Links to Dim_PaymentMethod
Fact_Sales	salesperson_id	INT	11	FK	Dim_Salesperson (salesperson_id)	Links to Dim_Salesperson
Fact_Sales	promotion_id	INT	11	FK	Dim_Promotion (promotion_id)	Links to Dim_Promotion
Fact_Sales	invoice_no	VARCHAR	30	-	-	Unique transaction number (degenerate dimension)
Fact_Sales	sales_quantity	INT	11	-	-	Units sold

Fact_Sales	discount_amount	DECIMAL	10,2	-	-	Total discount applied to the sale
Fact_Sales	sales_price	DECIMAL	10,2	-	-	Unit selling price after discount
Fact_Sales	sales_total	DECIMAL	12,2	-	-	Total revenue (sales_price * sales_quantity)

2) Fact_Service

Table	Attribute	Type	Format	Key	Reference	Description
Fact_Service	date_id	INT	11	FK	Dim_Date (date_id)	Links to Dim_Date
Fact_Service	car_id	INT	11	FK	Dim_Car (car_id)	Links to Dim_Car
Fact_Service	dealer_id	INT	11	FK	Dim_Dealership (dealer_id)	Links to Dim_Dealership
Fact_Service	service_type_id	INT	11	FK	Dim_ServiceType	Links to Dim_ServiceType
Fact_Service	customer_id	INT	11	FK	Dim_Customer (customer_id)	Links to Dim_Customer
Fact_Service	technician_id	INT	11	FK	Dim_Technician (technician_id)	Links to Dim_Technician
Fact_Service	payment_id	INT	11	FK	Dim_PaymentMethod (payment_id)	Links to Dim_PaymentMethod
Fact_Service	service_price	DECIMAL	10,2	-	-	Total service charge before discount
Fact_Service	downtime_days	INT	11	-	-	Number of days the car stayed in service
Fact_Service	warranty_flag	CHAR(1)	Y, N	-	-	Indicates whether the service is under warranty (Yes/No)
Fact_Service	discount_amount	DECIMAL	10,2	-	-	Total discount applied to the service charge
Fact_Service	service_in_date_id	DATE	YYYY-MM-DD	-	Dim_Date(date_id)	Service start date

Fact_Service	service_out_date_id	DATE	YYYY-MM-DD	-	Dim_Date(date_id)	Service completion date
Fact_Service	status	VARCHAR	20			Service status (Pending, Completed, In Progress)

3) Fact_Inventory

Table	Attribute	Type	Format	Key	Reference	Description
Fact_Inventory	date_id	INT	11	FK	Dim_Date (date_id)	Links to Dim_Date
Fact_Inventory	car_id	INT	11	FK	Dim_Car(car_id)	Links to Dim_Car
Fact_Inventory	dealer_id	INT	11	FK	Dim_Dealership (dealer_id)	Links to Dim_Dealership
Fact_Inventory	supplier_id	INT	11	FK	Dim_Supplier	Links to Dim_Supplier
Fact_Inventory	location_id	INT	11	FK	Dim_Location (location_id)	Links to Dim_Location
Fact_Inventory	stock_on_hand	INT	11	-	-	Current stock balance on that date (units)
Fact_Inventory	stock_in	INT	11	-	-	Units received into inventory on that date
Fact_Inventory	stock_out	INT	11	-	-	Units removed from inventory on that date (sales/transfers)
Fact_Inventory	reserved_qty	INT	11	-	-	Units reserved/not available for sale
Fact_Inventory	available_qty	INT	11	-	-	Units available for sale (stock_on_hand – reserved_qty)

Dimension Tables

1) Dim_Date

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Date	date_id	INT	11	PK	-	Surrogate key for date
Dim_Date	full_date	DATE	YYYY-MM-DD	-	-	Actual calendar date
Dim_Date	day	INT	2	-	-	Day of month
Dim_Date	month	INT	2	-	-	Month number
Dim_Date	month_name	VARCHAR	20	-	-	Month name
Dim_Date	quarter	INT	1	-	-	Quarter number
Dim_Date	Year	INT	4	-	-	Year

2) Dim_Car

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Car	car_id	INT	11	PK	-	Surrogate key for car
Dim_Car	model	VARCHAR	50	-	-	Car model
Dim_Car	color	VARCHAR	50	-	-	Color of car
Dim_Car	year	INT	4	-	-	Manufacture year
Dim_Car	base_price	DECIMAL	10,2	-	-	Standard price before discount
Dim_Car	transmission	VARCHAR	20	-	-	Transmission type
Dim_Car	fuel_type	VARCHAR	20	-	-	Fuel type
Dim_Car	engine_size	DECIMAL	3,1	-	-	Engine capacity
Dim_Car	engine_type	VARCHAR	50	-	-	Engine configuration
Dim_Car	seats	INT	1	-	-	Number of seats
Dim_Car	doors	INT	1	-	-	Number of doors

3) Dim_Dealership

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Dealership	dealer_id	INT	11	PK	-	Surrogate key for dealership
Dim_Dealership	dealer_name	VARCHAR	100	-	-	Dealership name
Dim_Dealership	city	VARCHAR	50	-	-	City of dealership
Dim_Dealership	province	VARCHAR	50	-	-	Province/state of dealership

Dim_Dealership	region	VARCHAR	50	-	-	Geographic region
Dim_Dealership	status	VARCHAR	30	-	-	Operational status
Dim_Dealership	phone	VARCHAR	20	-	-	Dealership contact number
Dim_Dealership	manager_name	VARCHAR	50	-	-	Dealership manager name
Dim_Dealership	store_size	VARCHAR	10	-	-	Small, Medium, Large

4) Dim_Customer

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Customer	customer_id	INT	11	PK	-	Surrogate key for customer
Dim_Customer	name	VARCHAR	50	-	-	Full name of customer
Dim_Customer	gender	VARCHAR	50	-	-	Male, Female, Other
Dim_Customer	date_of_birth	DATE	YYYY-MM-DD	-	-	Customer date of birth (used to derive age group)
Dim_Customer	occupation	VARCHAR	50	-	-	Customer's occupation
Dim_Customer	income_band	VARCHAR	20	-	-	Income category (Low, Medium, High)
Dim_Customer	city	VARCHAR	20	-	-	Customer's city
Dim_Customer	province	VARCHAR	50	-	-	Customer's province
Dim_Customer	region	VARCHAR	50	-	-	Customer's region
Dim_Customer	customer_segment	VARCHAR	50	-	-	Retail, Corporate, etc.

5) Dim_Salesperson

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Salesperson	salesperson_id	INT	11	PK	-	Surrogate key for salesperson
Dim_Salesperson	name	VARCHAR	50	-	-	Full name of the salesperson
Dim_Salesperson	gender	VARCHAR	50	-	-	Gender of the salesperson
Dim_Salesperson	date_of_birth	DATE	YYYY-MM-DD	-	-	Birth date of the salesperson

Dim_Salesperson	position	VARCHAR	30	-	-	Job position or role
Dim_Salesperson	status	VARCHAR	20	-	-	Employment status
Dim_Salesperson	hire_date	DATE	YYYY-MM-DD	-	-	Date the salesperson was hired

6) Dim_PaymentMethod

Table	Attribute	Type	Format	Key	Reference	Description
Dim_PaymentMethod	payment_id	INT	11	PK	-	Surrogate key for payment
Dim_PaymentMethod	payment_method	VARCHAR	20	-	-	Type of payment (Cash, Credit Card, Bank Loan, Leasing)
Dim_PaymentMethod	bank_name	VARCHAR	50	-	-	Name of the bank or financial institution
Dim_PaymentMethod	installment_months	INT	2	-	-	Number of months for installment payments
Dim_PaymentMethod	interest_band	VARCHAR	10	-	-	Interest rate category (Low, Medium, High)

7) Dim_ServiceType

Table	Attribute	Type	Format	Key	Reference	Description
Dim_ServiceType	service_type_id	INT	11	PK	-	Surrogate key for service type
Dim_ServiceType	service_type_name	VARCHAR	50	-	-	Name of the service (Oil Change, Engine Repair)
Dim_ServiceType	category	VARCHAR	20	-	-	Service category (Maintenance, Repair)
Dim_ServiceType	priority_level	VARCHAR	10	-	-	Priority classification (Low, Medium, High)

8) Dim_Technician

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Technician	technician_id	INT	11	PK	-	Surrogate key for technician
Dim_Technician	technician_name	VARCHAR	50	-	-	Full name of the technician
Dim_Technician	gender	VARCHAR	50	-	-	Gender of the technician
Dim_Technician	position	VARCHAR	50	-	-	Job role or specialization
Dim_Technician	status	VARCHAR	20	-	-	Employment status (Active, On Leave)

9) Dim_Location

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Location	location_id	INT	11	PK	-	Surrogate key for location
Dim_Location	location_name	VARCHAR	50	-	-	Name of the location
Dim_Location	location_type	VARCHAR	50	-	-	Type of location
Dim_Location	city	VARCHAR	50	-	-	City where the location is situated
Dim_Location	province	VARCHAR	50	-	-	Province of the location

10) Dim_Supplier

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Supplier	supplier_id	INT	11	PK	-	Surrogate key for supplier
Dim_Supplier	supplier_name	VARCHAR	50	-	-	Name of the supplier company
Dim_Supplier	contact_name	VARCHAR	50	-	-	Main contact person
Dim_Supplier	city	VARCHAR	50	-	-	City of the supplier
Dim_Supplier	province	VARCHAR	50	-	-	Province of the supplier
Dim_Supplier	region	VARCHAR	50	-	-	Geographic region
Dim_Supplier	country	VARCHAR	50	-	-	Country of the supplier

Dim_Supplier	status	VARCHAR	20	-	-	Supplier status
Dim_Supplier	email	VARCHAR	50	-	-	Supplier contact email
Dim_Supplier	phone	VARCHAR	20	-	-	Supplier contact phone number
Dim_Supplier	lead_time_days	INT	2	-	-	Average delivery lead time in days

11) Dim_Promotion

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Promotion	promotion_id	INT	11	PK	-	Surrogate key for promotion
Dim_Promotion	promo_name	VARCHAR	50	-	-	Promotion campaign name
Dim_Promotion	promo_type	VARCHAR	30	-	-	Type of promotion (Seasonal, Clearance, etc.)
Dim_Promotion	start_date	DATE	YYYY-MM-DD	-	-	Promotion start date
Dim_Promotion	end_date	DATE	YYYY-MM-DD	-	-	Promotion end date
Dim_Promotion	discount_type	VARCHAR	30	-	-	Discount method (Percent, Fixed amount, etc.)

4. Data Preparation

The data preparation process was conducted using Power Query Editor in Power BI Desktop. All 14 CSV files were imported and processed through the following steps to ensure data quality and consistency before loading into the data model.

Step 1: Data Import

All 14 CSV files were imported into Power BI Desktop via the Get Data → Text/CSV function. Each file was opened in Power Query Editor rather than loaded directly, to allow data cleaning and transformation to be performed prior to loading. The 14 files consist of three fact tables (Sales, Service, and Inventory) and eleven dimension tables (Car, Customer, Date, Dealership, Location, PaymentMethod, Promotion, Salesperson, Supplier, ServiceType, and Technician).

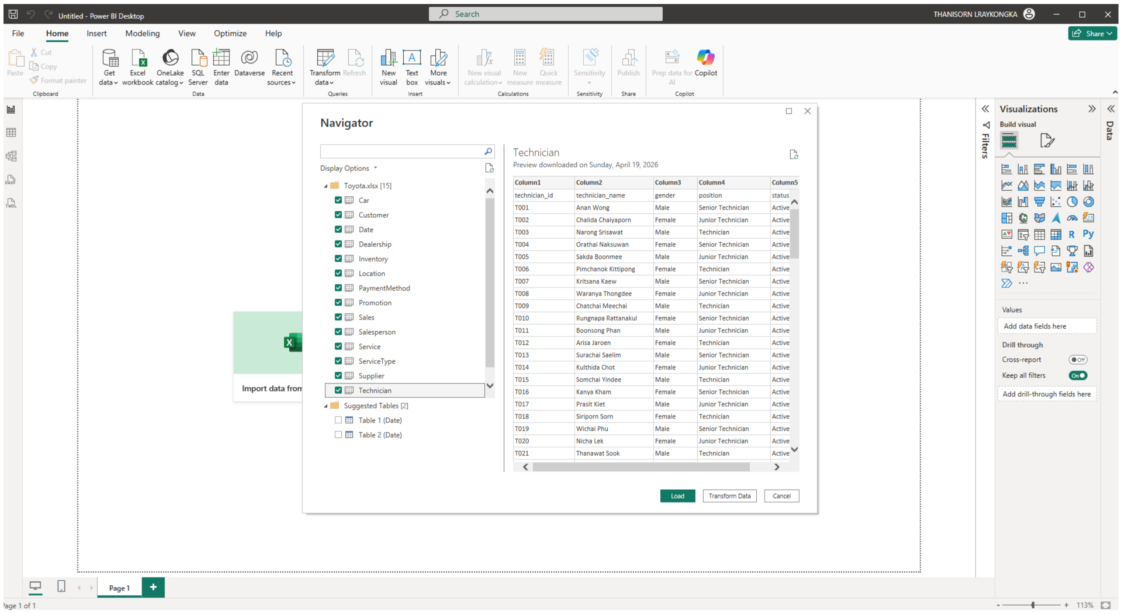


Figure 1: Importing and Preparing Raw Data Files

Step 2: Header Adjustment

After importing the CSV files into Power BI, all datasets were opened in Power Query Editor for further transformation. It was observed that some tables contained generic column names such as Column1, Column2, and so on, due to the absence of predefined headers during import.

To resolve this issue, the "Use First Row as Headers" function was applied to each affected table. This operation replaced the generic column names with the actual attribute names from the first row of the dataset, such as technician_id, technician_name, gender, position, and status in the Technician table.

This step ensured that all columns were properly labeled, improving data readability and enabling accurate data transformation and modeling in subsequent steps.

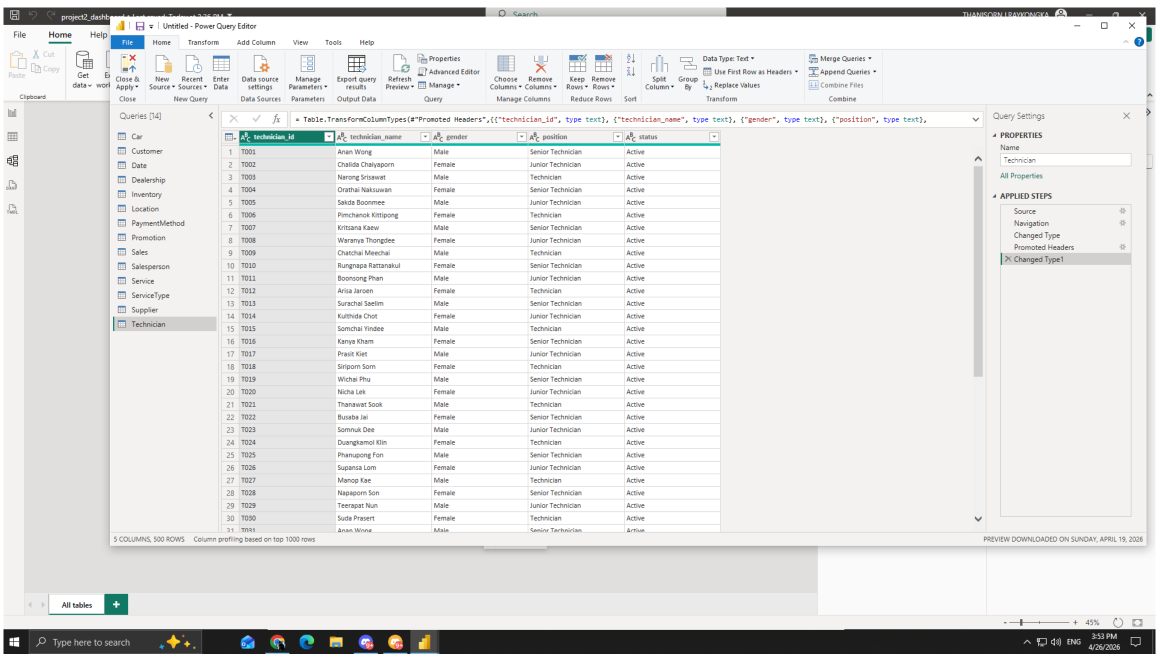


Figure 2: Applying "Use First Row as Headers"

Step 3: Data Type Correction

Upon importing the files, Power BI automatically detected data types for each column. However, several columns were assigned incorrect types and required manual correction to ensure accurate calculations and relationships. The corrections applied were as follows.

In the fact tables, all foreign key columns such as car_id, dealer_id, customer_id and related ID fields were verified and set to Text, except for data_id was set to Whole Number. Numeric measure columns such as sales_total, sales_price, and discount_amount in Fact_Sales, as well as service_price and discount_amount in Fact_Service, were set to Decimal Number. Count-based columns such as sales_quantity, stock_on_hand, stock_in, stock_out, reserved_qty, available_qty, and downtime_days were set to Whole Number. Only in date columns including full_date in Dim_Date, and service_in_date and service_out_date in Fact_Service were set as Text, therefore they were needed to be changed to Date type as refer to the Figure 3 and 4.

In the dimension tables, all primary key columns were set to Whole Number, and all descriptive text columns such as model, region, status, payment_method, and service_type_name were confirmed as Text type. The base_price column in Dim_Car and lead_time_days in Dim_Supplier were set to Decimal Number and Whole Number respectively.

When Power BI prompted whether to Replace current or Add new step during type conversion, the Replace current option was selected throughout to avoid redundant transformation steps and keep the Applied Steps list clean as shown in Figure 5.

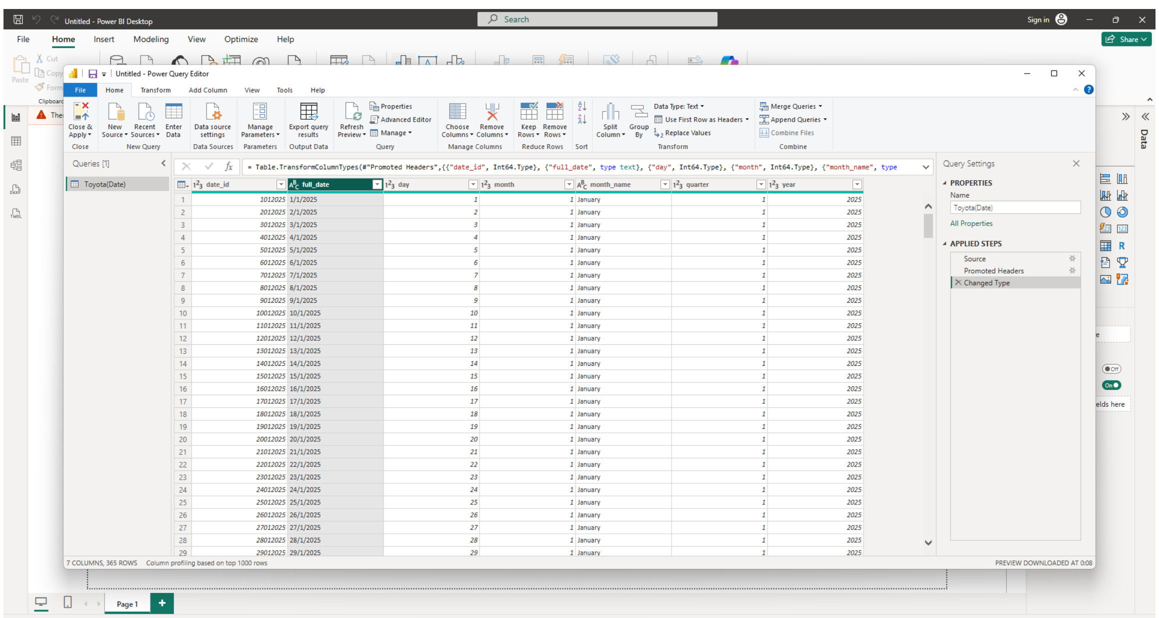


Figure 3: Identification of Inconsistent Date Formats in Raw Data

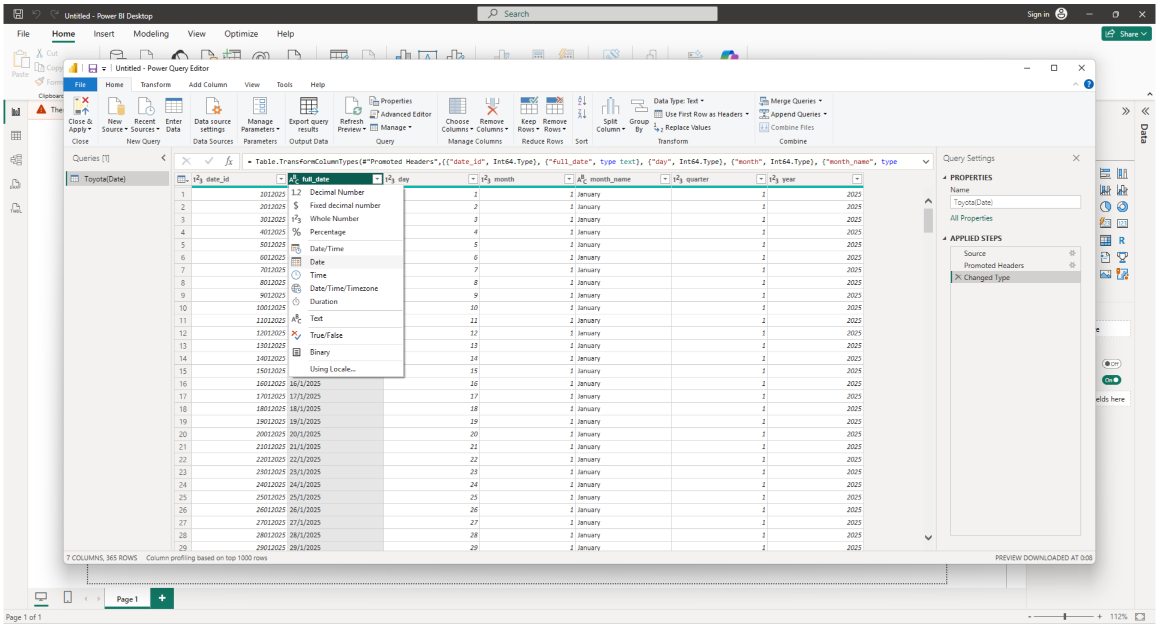


Figure 4: Reformatting Date Attributes via Power Query Editor

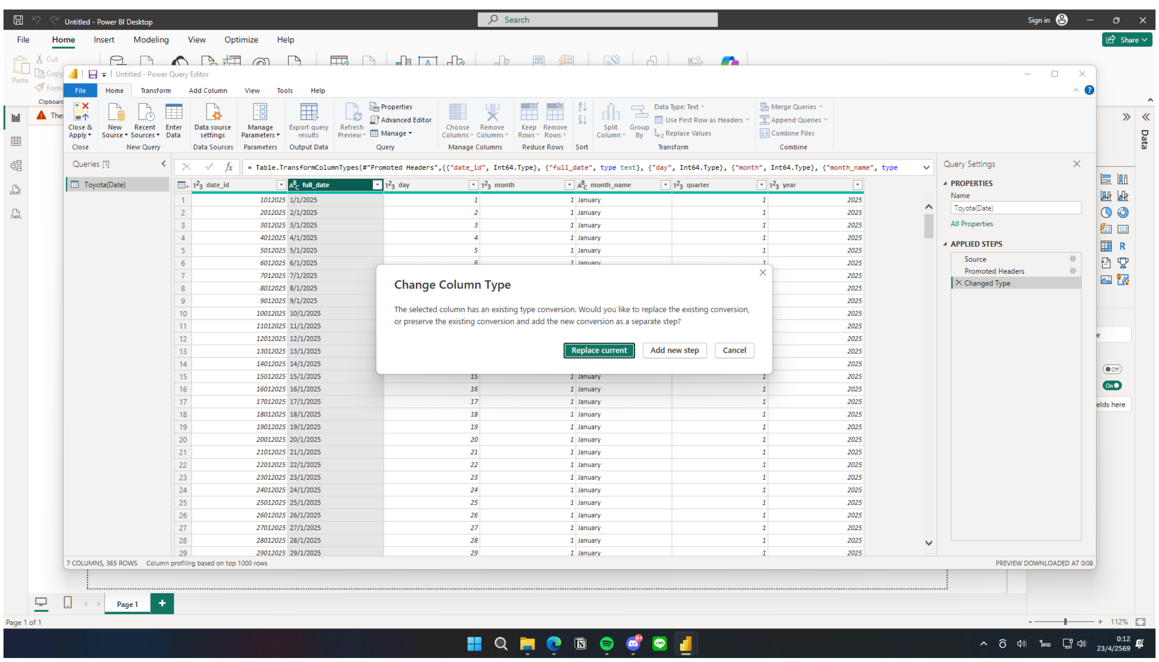


Figure 5: Validation of Standardized Date Formats

Step 4: Text Cleaning

To ensure consistency in text-based columns, the Trim and Clean functions were applied to all relevant text columns across all tables. This step removed leading and trailing whitespace as well as invisible characters that may have been introduced during data generation. Columns cleaned include names, region labels, status fields, and category values. This was performed by selecting each text column, then navigating to Transform → Format → Trim and Transform → Format → Clean as shown in Figure 6.

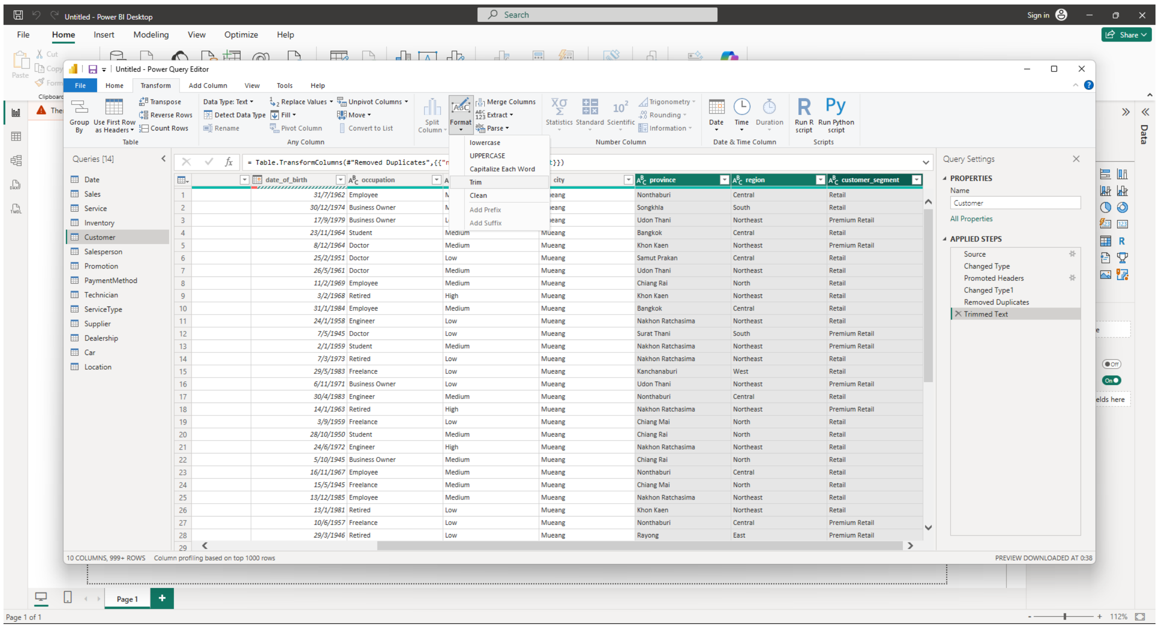


Figure 6: Applying the Trim Function to Remove Leading/Trailing Whitespace

Step 5: Remove Duplicates
Duplicate records were checked and removed for all dimension tables to ensure each primary key value appeared only once. This was done by selecting the primary key column of each dimension table and applying Remove Duplicates to maintain data integrity across relationships as shown in Figure 7.

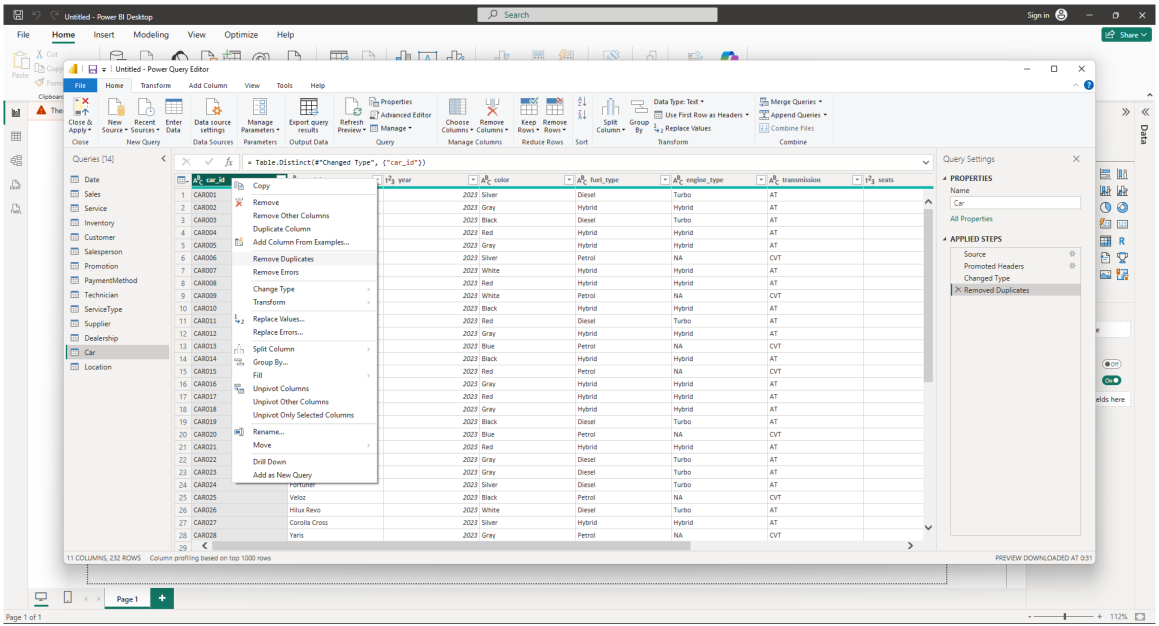


Figure 7: Executing 'Remove Duplicates' on Primary Key Attributes

Step 6: Data Loading and Modeling

After all cleaning and transformation steps were completed, the data was loaded into Power BI by selecting Close & Apply. Relationships between fact tables and dimension tables were then established in the Model View using a star schema structure. All relationships were configured as Many-to-One from the fact table to the respective dimension table, based on matching key columns. A total of three-star schemas were created, one for each fact table: Fact_Sales, Fact_Service, and Fact_Inventory refer to Figure 8.

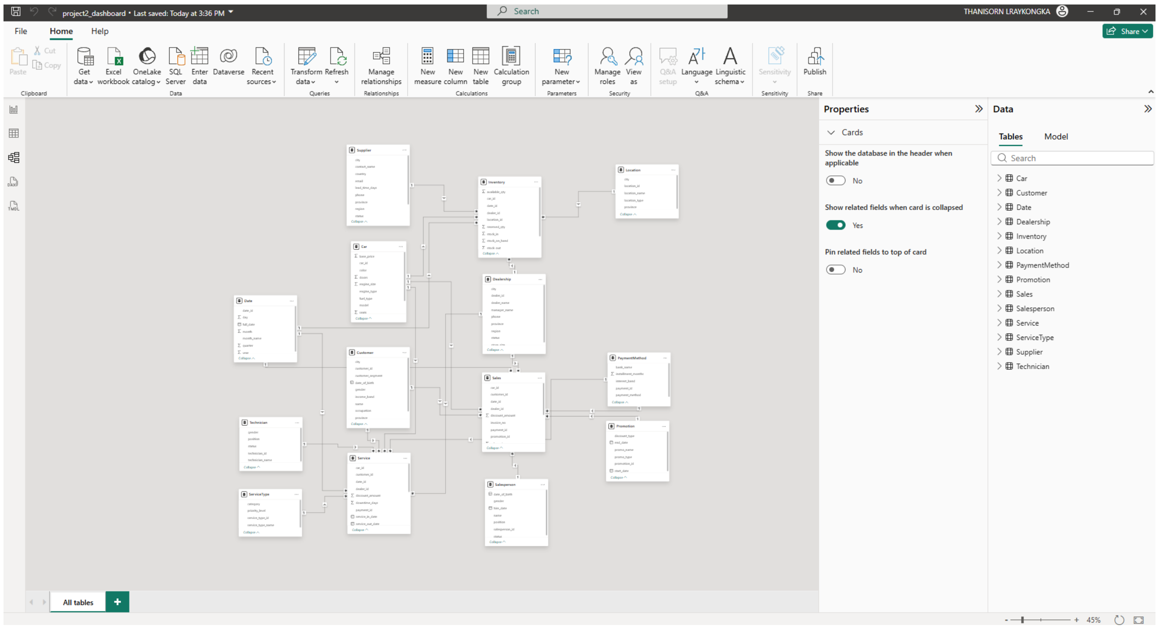


Figure 8: Toyota Data Model

5. Business Intelligence Implementation and Analytics

5.1 BI Tool Selection

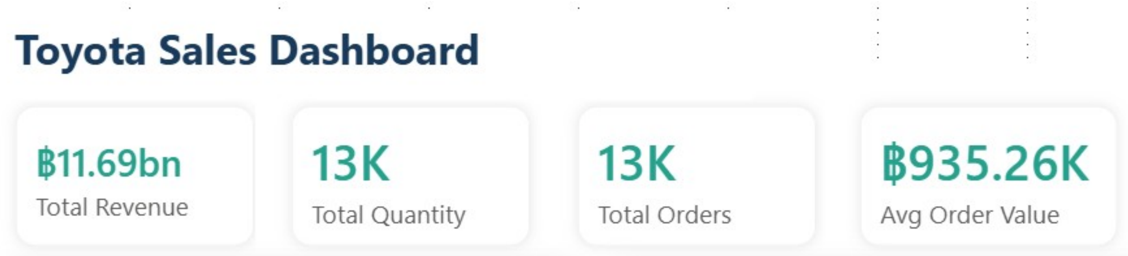
Power BI was selected as the Business Intelligence tool for this project due to its strong data visualization capabilities and intuitive user interface. It supports seamless integration with multiple data sources and enables the development of interactive dashboards for efficient data exploration.

Furthermore, Power BI provides a wide range of visualization options, allowing clear presentation of trends, patterns, and key performance indicators (KPIs). Its filtering and drill-down functionalities enable users to analyze data at multiple levels of detail, enhancing analytical flexibility.

In this project, Power BI is used to visualize sales, service, and inventory data, allowing users to explore performance across different dimensions such as time, region, and product. This supports better understanding of business operations and facilitates more informed decision-making.

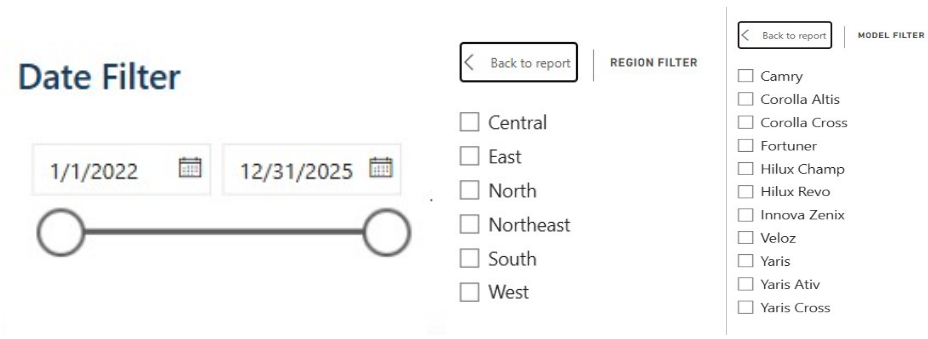
5.2 Analysis and Visualization Reports

Sales Overview



KPI Cards

Four KPI cards are displayed at the top of the dashboard to provide an immediate summary of overall sales performance. These include Total Revenue, Total Quantity, Total Orders, and Average Order Value. These metrics allow managers to quickly assess business performance without navigating through individual charts.



Filters

These filters allow users to interactively explore the data by selecting specific time periods, regions, and car models, enabling flexible and customized analysis.

Total Revenue by Year, Quarter, Month

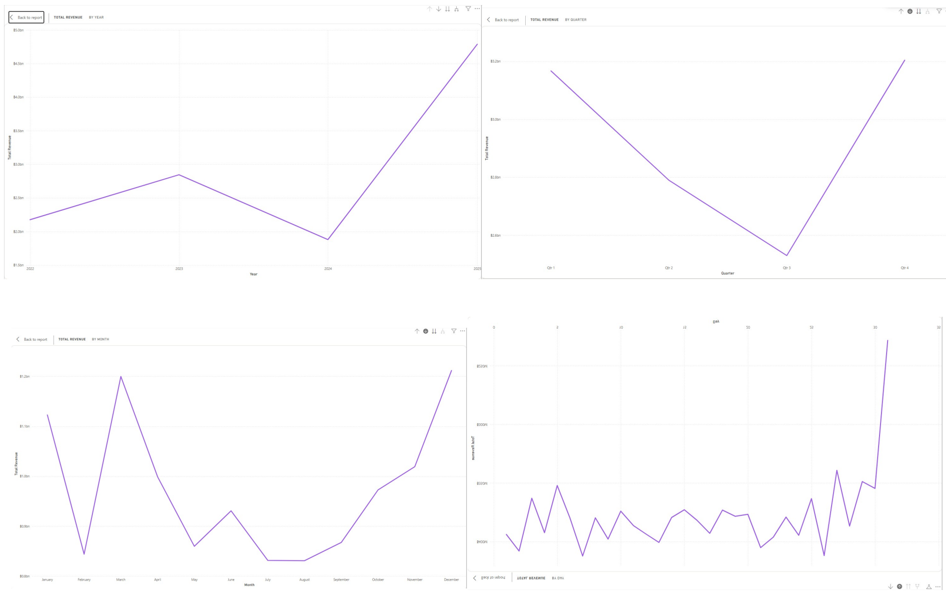


Figure 9: Line Chart of Total Revenue in Fact Sales

Description

The chart in Figure 9 presents total revenue over time, allowing users to analyze trends and compare performance across different periods. It supports drill-down and drill-up (year, quarter, month, day) for more detailed analysis.

Insight

Revenue shows a fluctuating pattern, with growth in 2023, a decline in 2024, and a strong rebound in 2025. The sharp recovery after the decline suggests that the drop in 2024 may not be structural but rather linked to temporary factors such as reduced promotional intensity, weaker regional performance, or lower demand for specific car models during certain periods. This indicates a temporary performance issue, and Toyota should diversify its model portfolio to reduce reliance on top-selling models and mitigate revenue concentration risk.

Recommendation

Toyota should use drill-down analysis to identify specific months or quarters responsible for the decline and evaluate related factors such as promotions, regions, and product performance. Successful strategies from 2025 should be replicated to stabilize revenue growth.

Revenue by Car Model

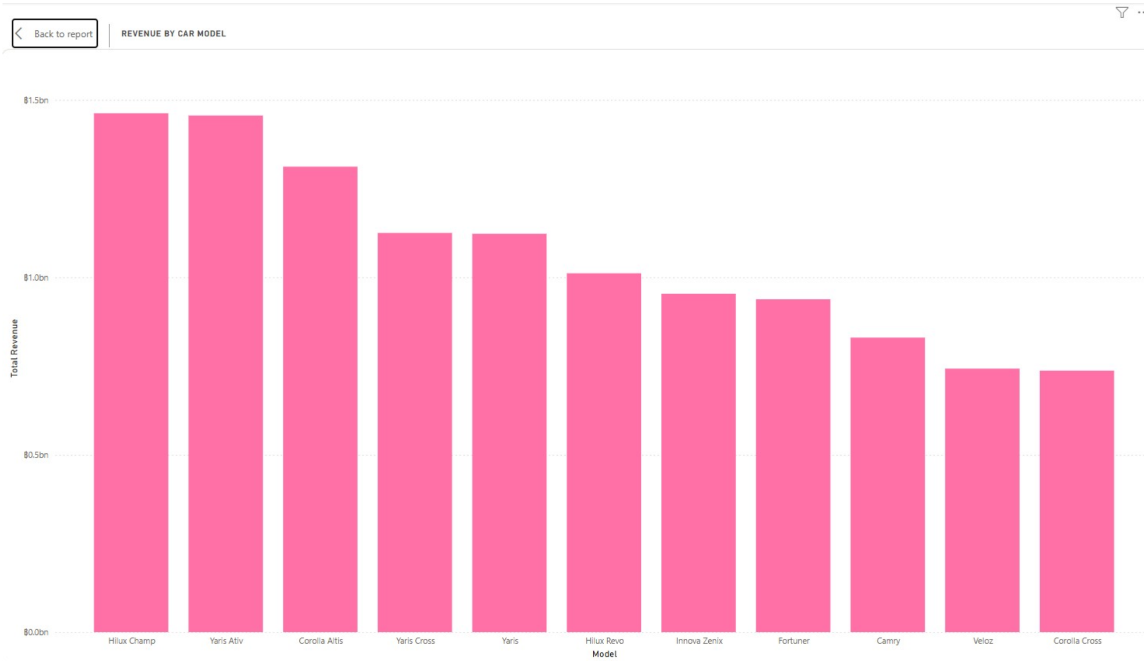


Figure 10: Bar Chart of Total Revenue of Car Models in Fact Sales

Description

This chart compares total revenue generated by each Toyota vehicle model, helping identify which products drive the most business value as shown in Figure 10.

Insight

Hilux Champ generates the highest revenue among all models, while smaller models such as Yaris contribute less. This reflects customer preference for utility and pickup vehicles. Furthermore, revenue is concentrated in a few car models, indicating strong customer preference for specific products. This suggests that demand is not evenly distributed and may depend on factors such as price range, functionality, or brand perception. This suggests that sales performance is highly concentrated, and expanding marketing efforts to underperforming regions and promoting a broader range of models could improve overall revenue stability.

Recommendation

Toyota should prioritize high-performing models in marketing and inventory planning while revising strategies for lower-performing models to improve overall product portfolio balance.

Top 5 Dealerships by Revenue

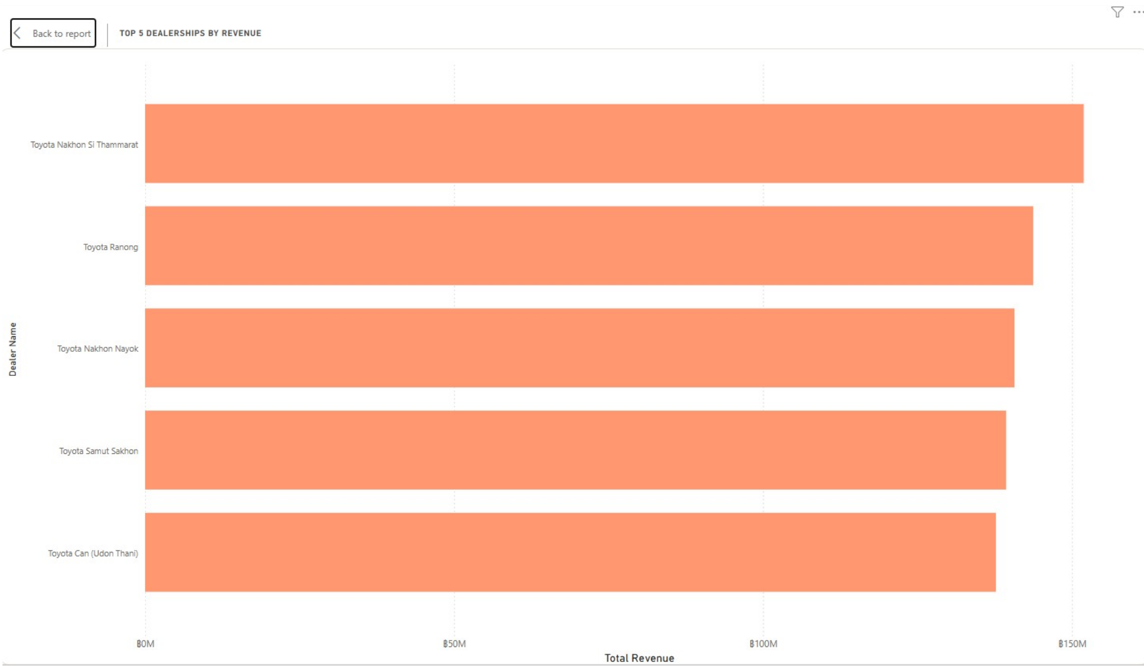


Figure 11: Bar Chart of Top 5 Dealership by Total Revenue

Description

This bar chart referring in Figure 11 ranks the top five dealerships based on total revenue, highlighting key contributors to overall performance.

Insight

The leading dealership generates noticeably higher revenue than others, while the remaining dealerships perform at similar levels. This indicates that revenue is concentrated in a few key locations, suggesting differences in management effectiveness, customer base, or local market conditions.

Recommendation

Toyota should analyze operational strategies from top-performing dealerships and replicate them in lower-performing locations to improve overall consistency and maximize revenue potential.

Top 5 Salesperson by Revenue

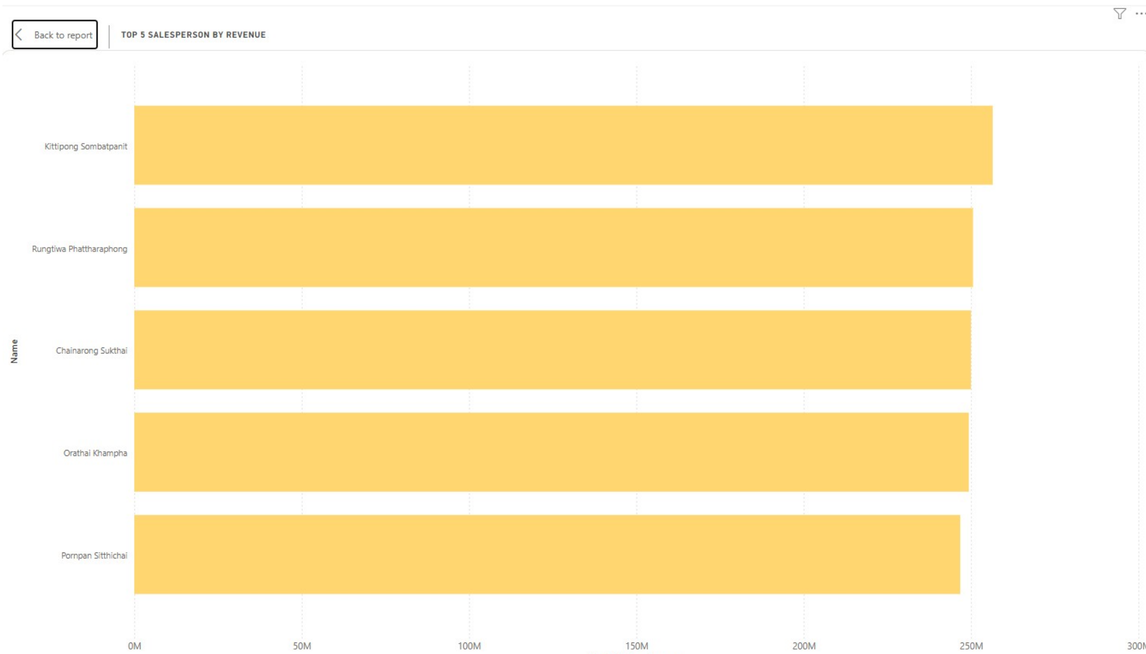


Figure 12: Bar Chart of Top 5 Salesperson by Total Revenue

Description

This bar chart compares revenue generated by the top five salespersons, enabling performance evaluation across individuals as seen on Figure 12.

Insight

Revenue is concentrated among a small number of top-performing salespersons, indicating uneven productivity within the sales team. This suggests that performance may depend on individual skills, experience, or customer networks rather than consistent organizational capability.

Recommendation

Toyota should standardize best practices from top performers and implement training programs to improve the overall productivity and consistency of the sales team.

Sales by Region

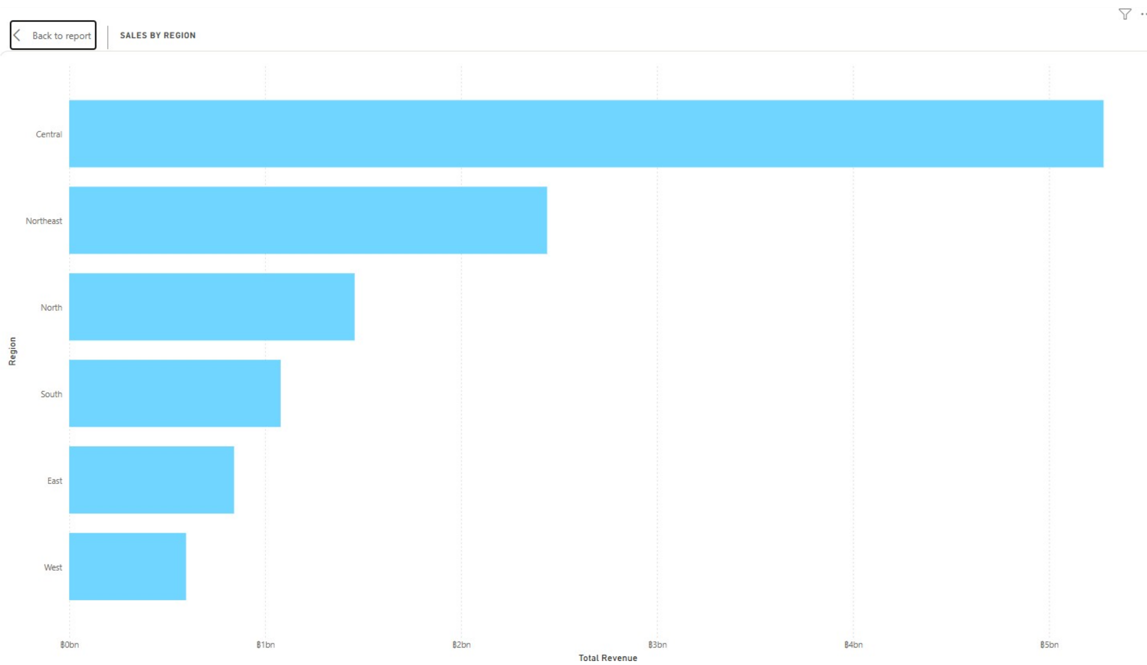


Figure 13: Bar Chart of Top 5 Dealership by Total Revenue

Description

This bar chart in Figure 13 compares total revenue across geographic regions, providing a view of regional market strength and distribution of sales performance.

Insight

Significant variation in revenue across regions suggests uneven market performance. This may reflect differences in customer demand, economic conditions, or dealership effectiveness across locations.

Recommendation

Toyota should tailor marketing and operational strategies to each region, focusing on improving performance in underperforming areas while maintaining strong positions in high-performing regions.

Top 10 Revenue by Promotion

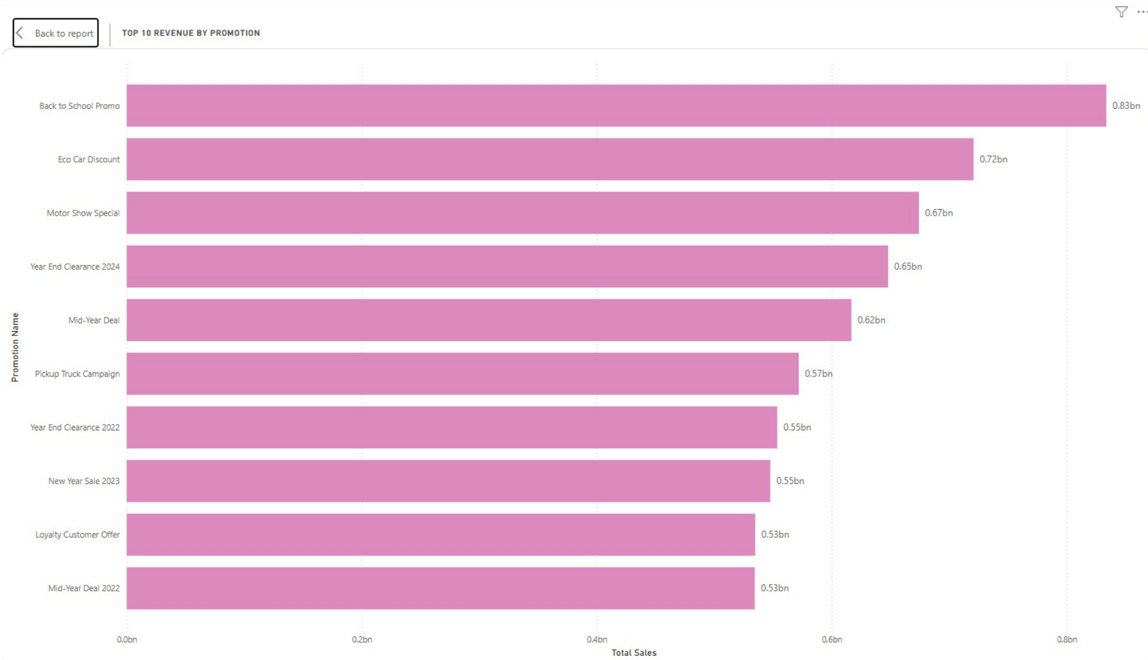


Figure 14: Bar Chart of Top 10 Revenue by Promotion

Description

The chart in Figure 14 valuates the revenue impact of each promotional campaign, providing insight into which campaigns are most effective in driving sales.

Insight

A small number of promotions generate significantly higher revenue than others, indicating that customer response is highly sensitive to specific campaign types. This suggests that not all promotions are equally effective, and certain campaigns likely align better with customer demand or timing.

Recommendation

Toyota should analyze the characteristics of top-performing promotions (e.g., timing, discount type, target segment) and apply these insights to optimize future campaigns while reducing reliance on less effective ones.

Total Revenue by Payment Method

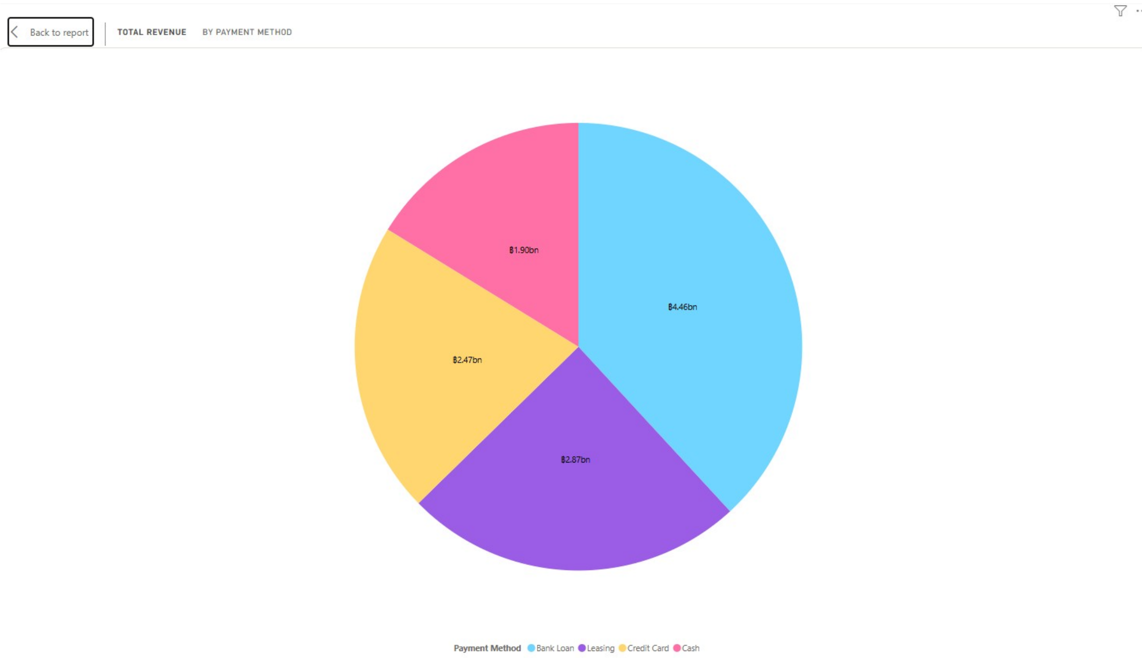


Figure 15: Pie Chart of Total Revenue of Payment Method

Description

This chart in Figure 15 illustrates how revenue is distributed across payment methods such as Cash, Bank Loan, Credit Card, and Leasing.

Insight

Financing options such as bank loans and leasing dominate revenue contribution, while cash payments are minimal. This indicates that vehicle purchases are highly dependent on financing accessibility, and customer purchasing power is closely tied to credit availability.

Recommendation

Toyota should strengthen partnerships with financial institutions and offer flexible financing plans to support customer purchasing decisions and sustain revenue growth.

Revenue by Customer Segment

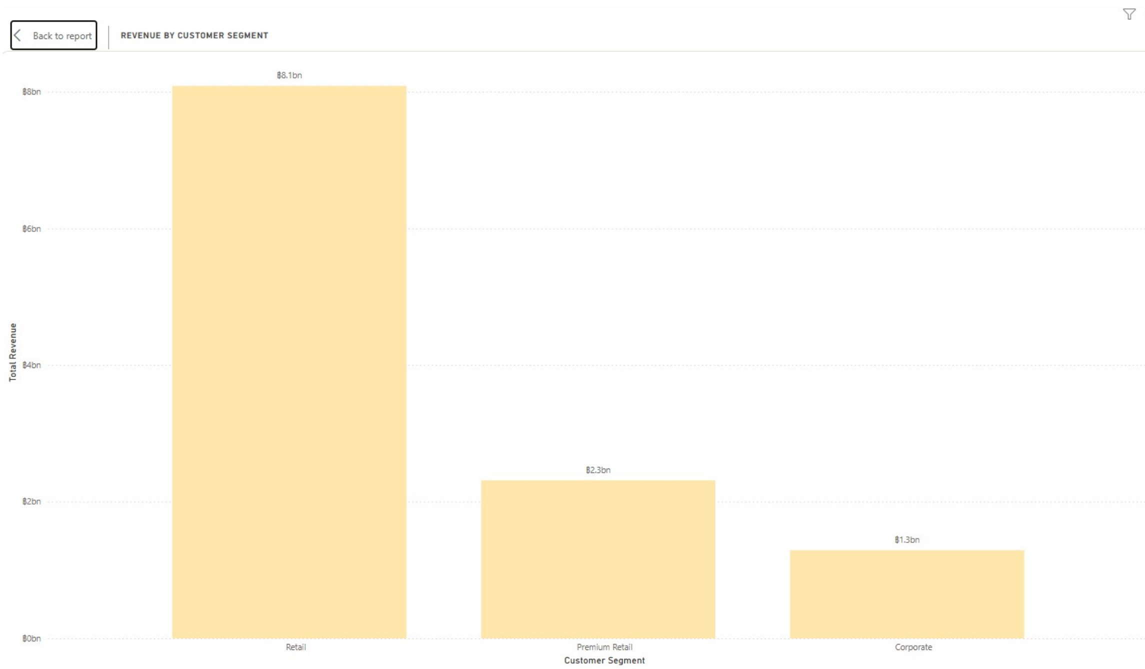


Figure 16: Bar Chart of Revenue by Customer Segment

Description

This bar chart compares total revenue across different customer segments, enabling identification of key contributing groups as shown in Figure 16.

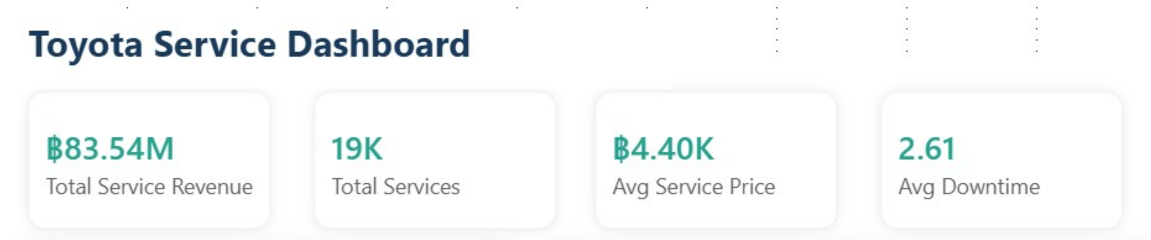
Insight

Retail and Premium Retail segments generate the majority of revenue, while Corporate contributes significantly less. This indicates that revenue is heavily dependent on individual customers, suggesting limited penetration in the corporate market and potential underutilization of bulk or fleet sales opportunities.

Recommendation

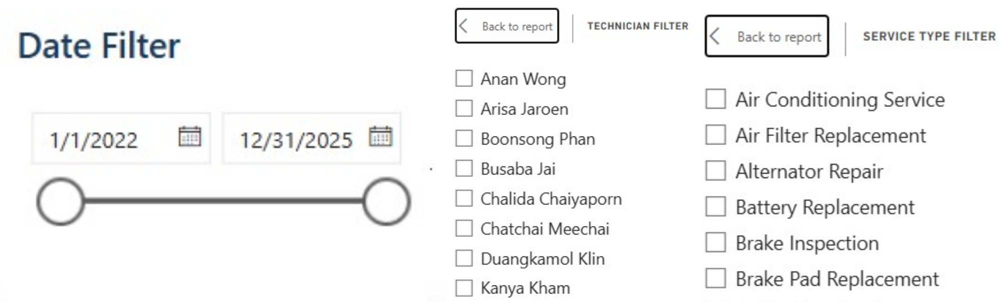
Toyota should strengthen engagement with high-value retail customers while developing targeted strategies, such as corporate partnerships or fleet sales programs, to expand the corporate segment.

Service Overview



KPI Cards

These KPI cards display key service performance metrics, including Total Service Revenue, Total Services, Average Service Price, and Average Downtime, providing a quick overview of overall service operations.



Filters

These filters allow users to interactively explore service data by selecting specific time periods, technicians, and service types, enabling flexible and customized analysis.

Top 10 Revenue by Service Type

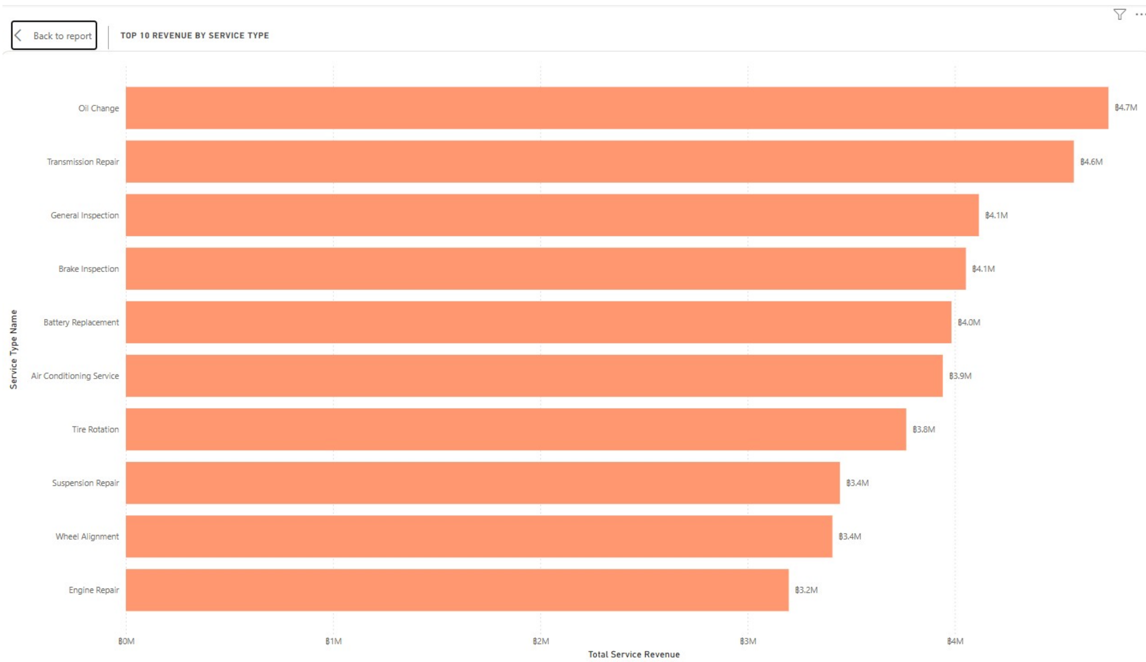


Figure 17: Bar Chart of Top 10 Revenue by Service Type

Description

This bar chart in Figure 17 displays the top 10 service types ranked by total service revenue, enabling comparison across different service categories and identifying key revenue contributors.

Insight

Revenue is concentrated in a few core services such as maintenance and repair (e.g., oil change, transmission repair), indicating that demand is driven by recurring and essential services. The relatively small gap between top service types suggests stable demand across multiple service categories rather than reliance on a single service.

Recommendation

Toyota should prioritize high-demand service types by ensuring resource availability and service efficiency, while also promoting complementary services to increase overall service revenue.

Top 5 Technician Performance (Revenue vs Avg Downtime)

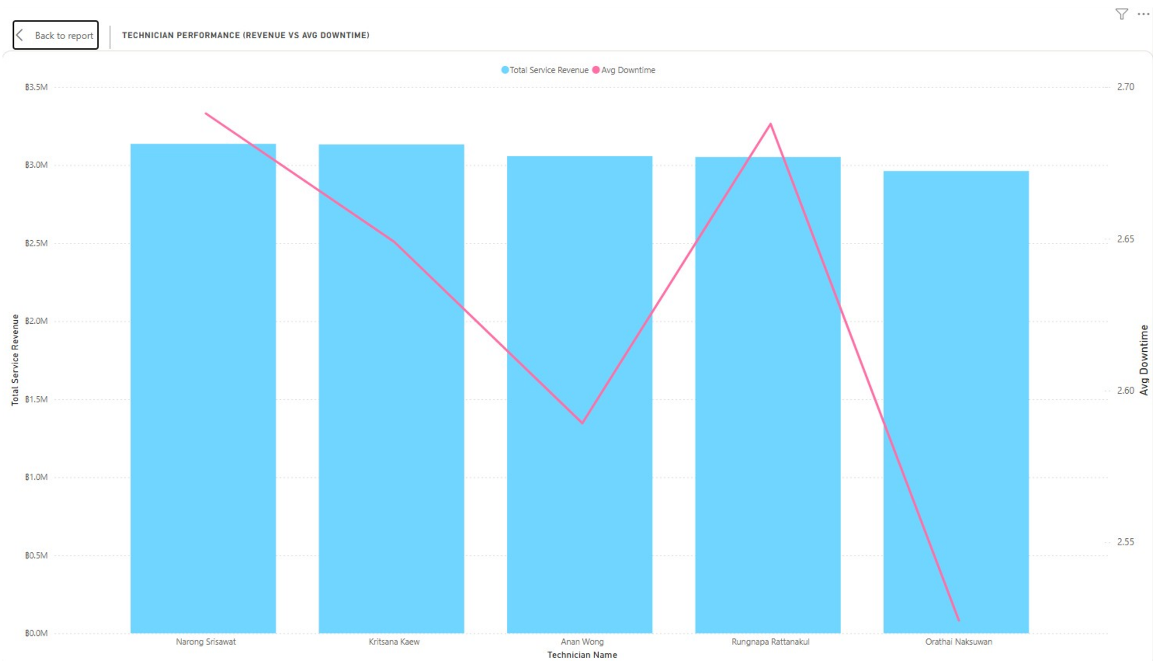


Figure 18: Bar Chart and Line Chart of Top 5 Technician Performance

Description

This combined chart in Figure 18 compares technician performance based on total service revenue (bar) and average downtime (line), enabling evaluation of both productivity and efficiency.

Insight

Some technicians generate high revenue while maintaining lower downtime, indicating strong efficiency. In contrast, others show higher downtime despite similar revenue levels, suggesting inconsistency in performance.

Recommendation

Toyota should use high-performing technicians as benchmarks and provide targeted training or process improvements to enhance overall technician efficiency.

Service Status Distribution

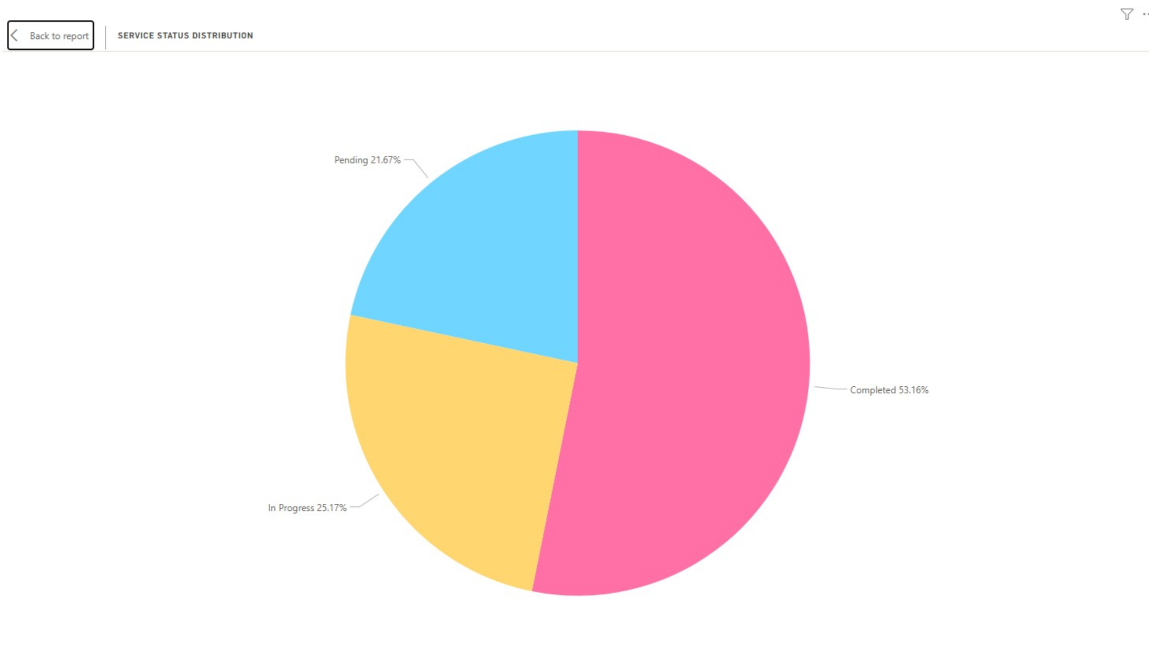


Figure 19: Pie Chart of Service Status Distribution

Description

This pie chart displays the proportion of service statuses, including completed, in-progress, and pending services, providing an overview of operational workflow as shown in Figure 19.

Insight

While the majority of services are completed, a noticeable portion remains in-progress and pending, indicating potential bottlenecks in service operations. This suggests that workflow efficiency may not be fully optimized.

Recommendation

Toyota should improve service process management and resource allocation to reduce pending and in-progress workloads, ensuring faster service completion.

Service Count by Warranty

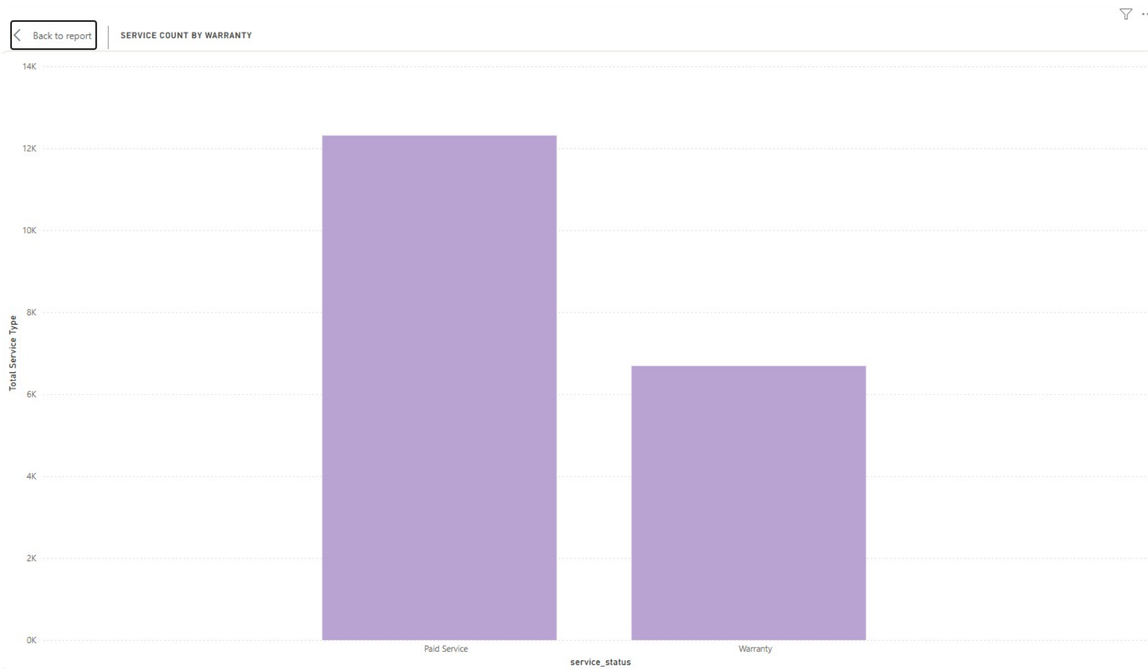


Figure 20: Bar Chart of Service Count by Warranty

Description

This bar chart referring in Figure 20 compares the number of services performed under paid service and warranty categories, highlighting differences in service type distribution.

Insight

Paid services significantly exceed warranty services, indicating that most service revenue is generated from non-warranty maintenance rather than manufacturer-covered repairs. This suggests strong reliance on after-sales service as a revenue stream. This highlights an opportunity for Toyota to further enhance customer retention strategies and develop service packages to maximize long-term revenue.

Recommendation

Toyota should continue strengthening after-sales service offerings and introduce service packages or maintenance plans to further increase paid service engagement.

Service Volume Trend

Figure 21: Line Chart of Service Volume Trend

Description
This line chart shows the trend of service volume over time, allowing analysis of changes in service demand across different periods—such as by year, quarter, month, or even day—using drill-down capabilities, as shown in Figure 21.

Insight
Service volume increases in 2023, drops in 2024, and rises again in 2025, indicating fluctuations in service demand. By drilling down to yearly, quarterly, monthly, or daily views, potential seasonal or operational factors influencing these shifts can be identified more precisely.

Recommendation
Toyota should analyze periods of decline by using time-based drill-down (year, quarter, month, day) to pinpoint causes at different granularities and implement strategies to maintain consistent service demand.

Service Frequency & Average Downtime by Car Model

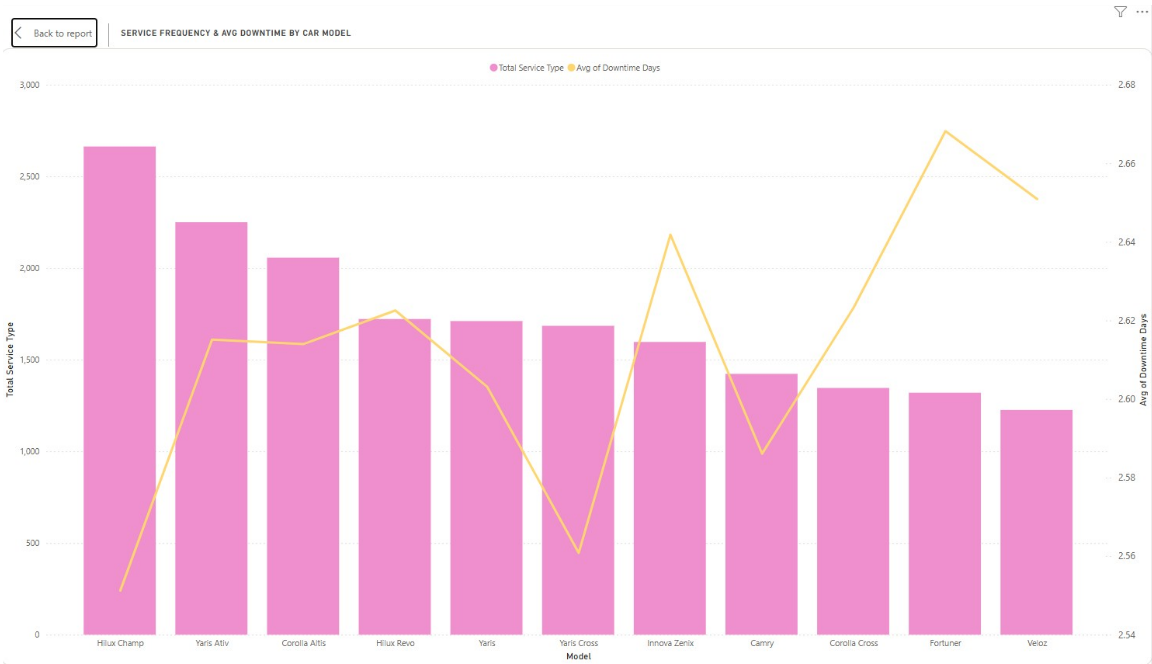


Figure 22: Bar Chart and Line Chart of Service Frequency and Average Downtime

Description

This combined chart in Figure 22 shows service frequency (bar) and average downtime (line) by car model, enabling comparison of service demand and operational efficiency across products.

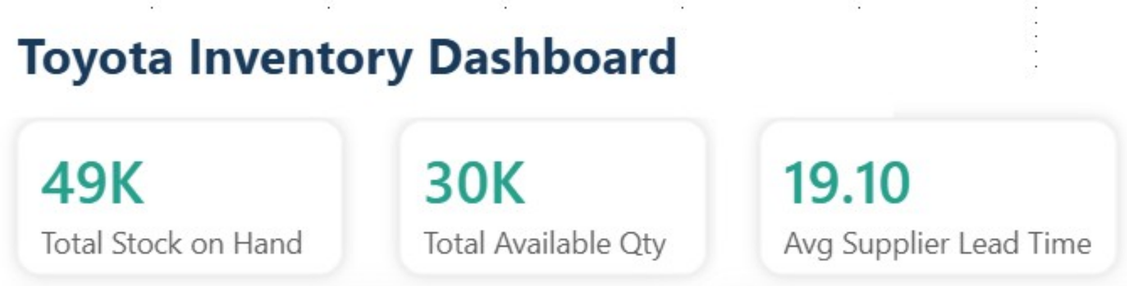
Insight

Some car models have high service frequency but relatively stable downtime, indicating consistent demand and efficient service handling. However, models with higher downtime despite lower frequency may indicate inefficiencies or more complex service requirements.

Recommendation

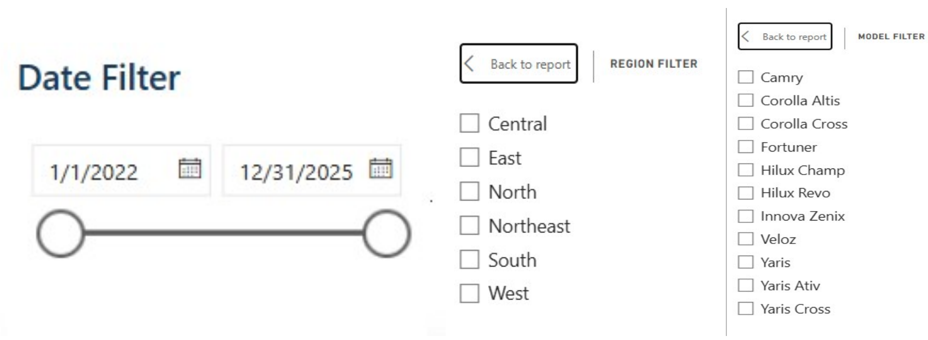
Toyota should investigate models with high downtime to improve service processes or technical support, while maintaining efficiency for high-frequency models to ensure customer satisfaction.

Inventory Overview



KPI Cards

These KPI cards display key inventory metrics, including Total Stock on Hand, Total Available Quantity, and Average Supplier Lead Time, providing a quick overview of inventory status and supply chain performance.



Filters

These filters allow users to interactively explore inventory data by selecting specific time periods, regions, and car models, enabling flexible and customized analysis.

Stock on Hand by Dealership

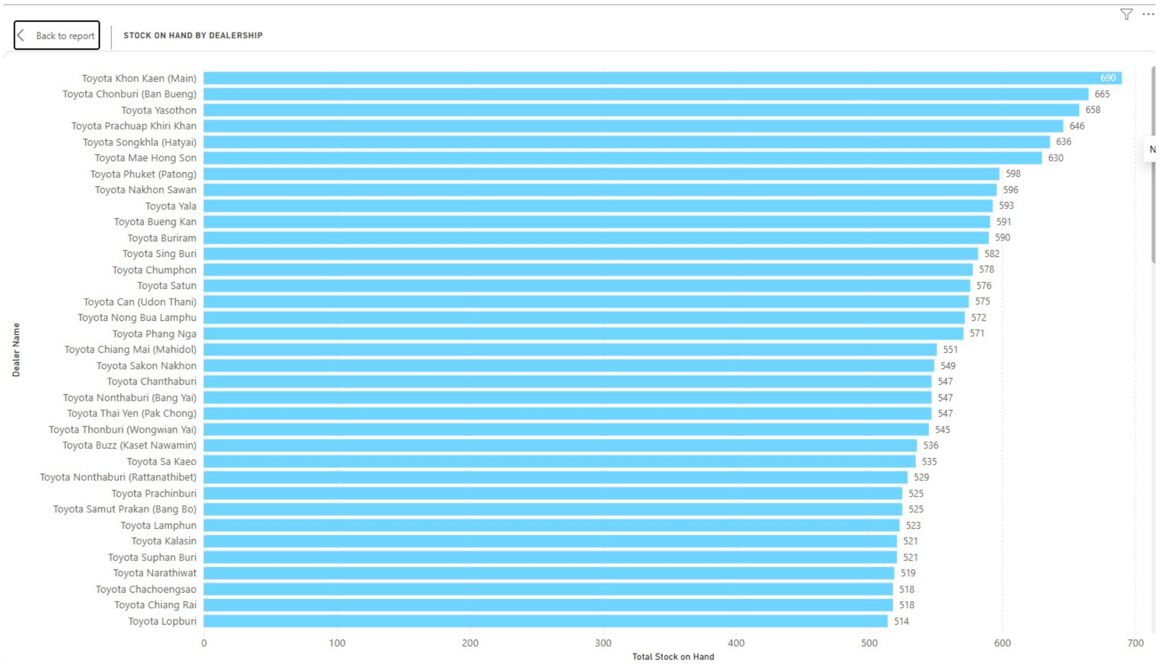


Figure 23: Bar Chart of Stock on Hand by Dealership

Description
This bar chart in Figure 23 shows the total stock on hand across different dealerships, enabling comparison of inventory levels between locations.

Insight

Inventory levels vary significantly across dealerships, with some locations holding much higher stock than others. This indicates an imbalance in inventory distribution, which may lead to overstock in some areas and potential stock shortages in others, impacting sales opportunities and increasing holding costs.

Recommendation
Toyota should optimize inventory allocation by redistributing stock based on regional demand patterns and sales performance. Additionally, implementing data-driven inventory planning and demand forecasting can help ensure more balanced stock levels and improve overall operational efficiency.

Average Supplier Lead Time (Days)

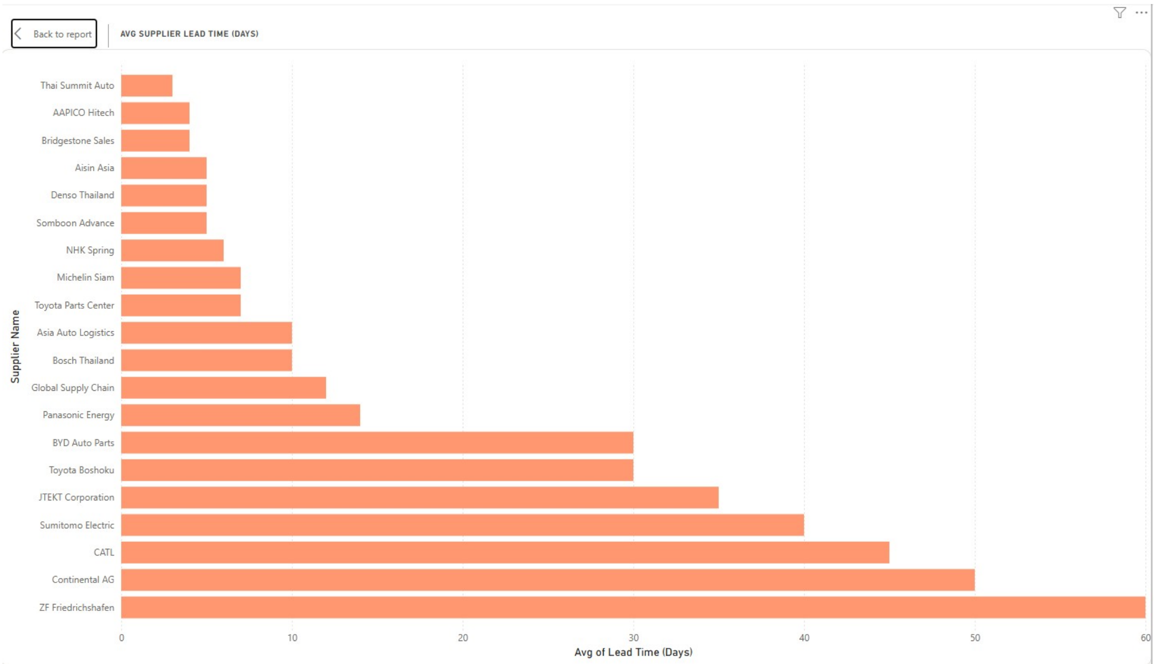


Figure 24: Bar Chart of Average Supplier Lead Time

Description

This bar chart in Figure 24 shows the average supplier lead time, enabling comparison of delivery performance across suppliers.

Insight

There is a significant variation in lead time between suppliers, with some taking considerably longer than others. This indicates potential risks in supply chain reliability and delays in inventory replenishment.

Recommendation

Toyota should prioritize suppliers with shorter lead times and improve coordination with slower suppliers to reduce delays and enhance supply chain efficiency.

Stock Turnover by Region

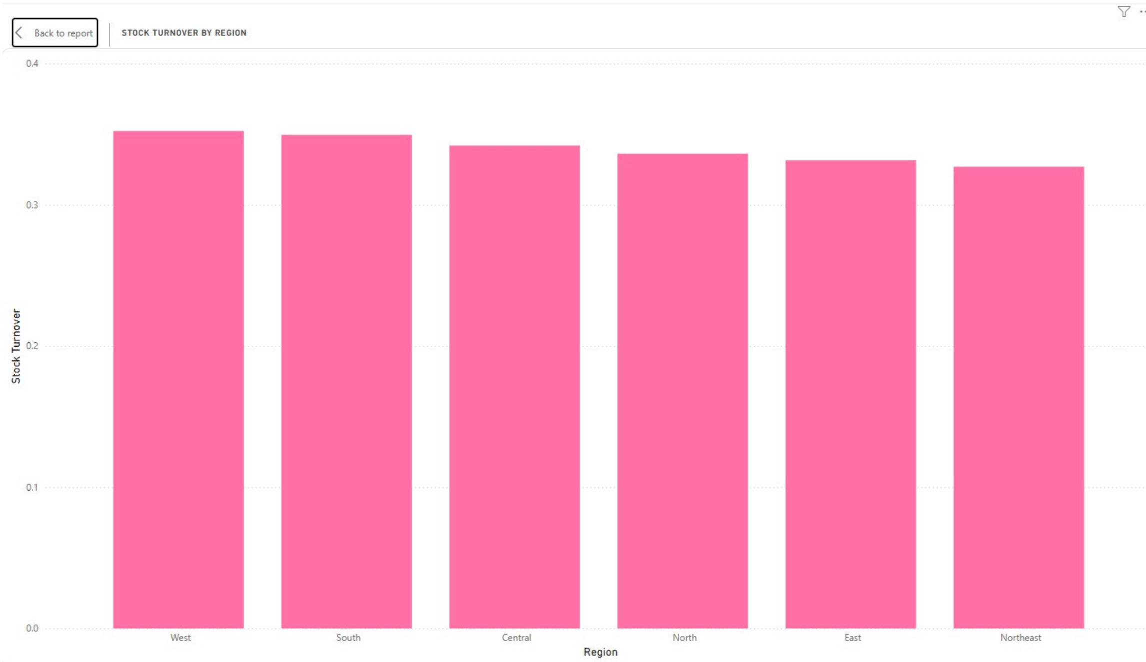


Figure 25: Bar Chart of Stock Turnover by Region

Description

This bar chart compares stock turnover rates across regions, indicating how efficiently inventory is being sold as shown in Figure 25.

Insight

Stock turnover rates are relatively similar across regions but show slight variations, suggesting moderate consistency with minor inefficiencies in certain areas. This indicates that while overall performance is stable, there is still room for optimization.

Recommendation

Toyota should analyze best-performing regions and apply their inventory management practices to underperforming areas to improve overall turnover efficiency.

Available vs Reserved Stock by Dealership

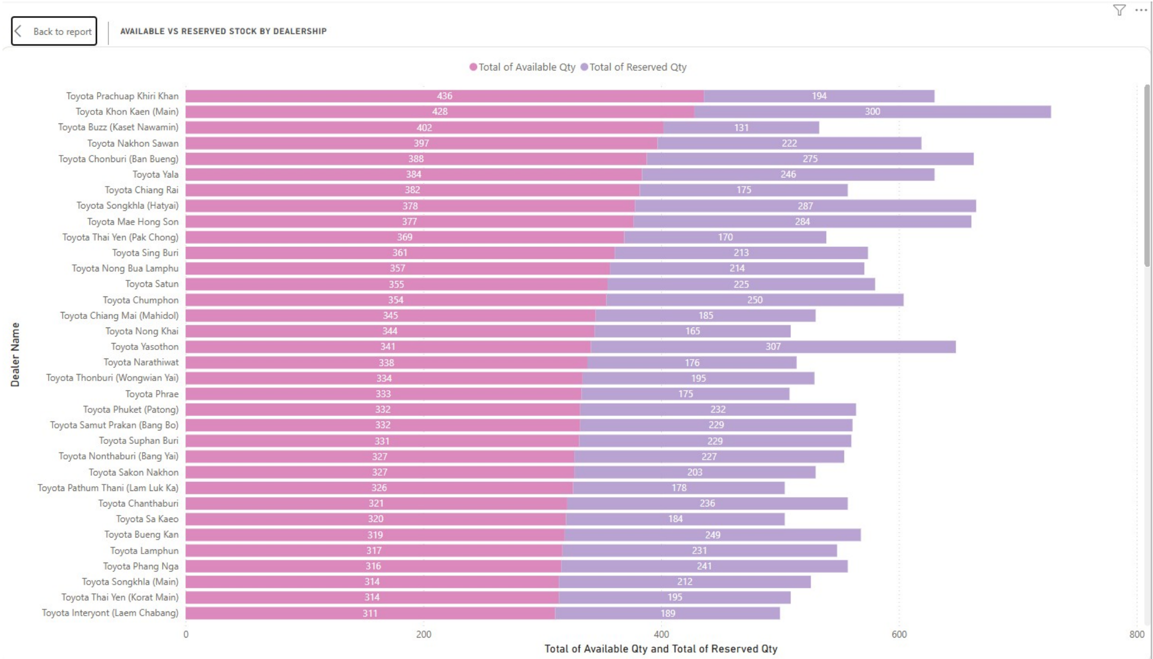


Figure 26: Bar Chart of Available vs Reserved Stock by Dealership

Description

This chart compares available stock and reserved stock across dealerships, enabling analysis of inventory utilization as seen in Figure 26.

Insight

Some dealerships show high available stock relative to reserved stock, indicating underutilization of inventory. In contrast, others have more balanced levels, suggesting better demand alignment.

Recommendation

Toyota should improve inventory utilization by aligning stock levels with customer demand and reservation patterns, ensuring efficient use of available inventory.

Stock on Hand by Car Model (Slow-moving)

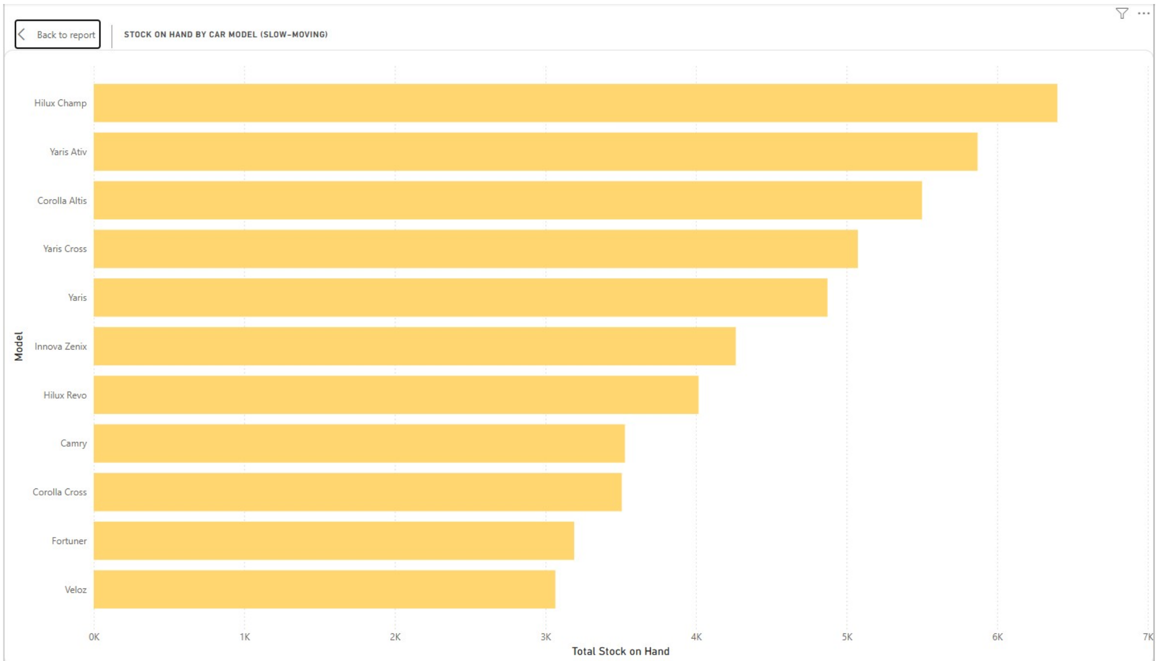


Figure 27: Bar Chart of Stock on Hand by Car Model (Slow-moving)

Description

This bar chart in Figure 27 displays stock levels of slow-moving car models, helping identify products with low turnover.

Insight

Certain car models have significantly higher stock levels despite being categorized as slow-moving, indicating potential overstock and low demand. This suggests inefficiencies in inventory planning and demand forecasting.

Recommendation

Toyota should reduce excess inventory for slow-moving models through targeted promotions, pricing adjustments, or production planning improvements.

5.3 Dashboard Design

Sales Dashboard

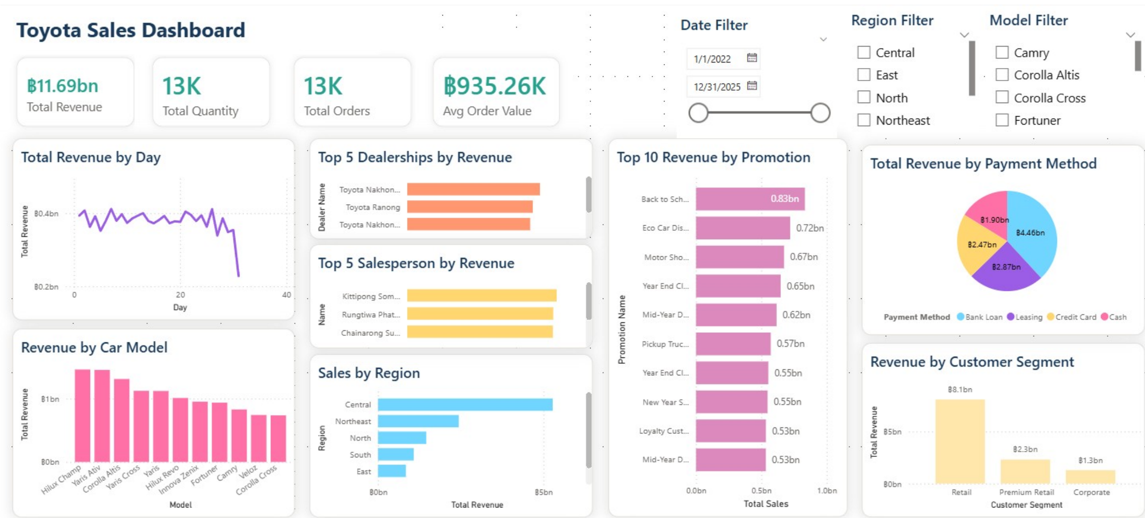


Figure 28: Toyota Sale Dashboard

The Sales Dashboard referring to Figure 28 is designed to provide a clear and interactive overview of business performance. The layout follows a **top-down structure**, where key performance indicators (KPIs), including Total Revenue, Total Orders, Total Quantity, and Average Order Value, are placed at the top for quick understanding.

Below the KPIs, multiple visualizations are arranged logically to support analysis. A **line chart** is used to display revenue trends over time, allowing users to easily observe changes across years. **Bar charts** are applied for comparisons such as revenue by region, car model, and dealership performance, as they effectively highlight differences between categories. A **pie chart** is used to represent payment method distribution, providing a clear view of proportion.

The dashboard includes **interactive filters** such as date, region, and model, enabling users to customize their view and explore data dynamically. The use of consistent colors and clear labeling enhances readability, while the structured layout ensures that users can navigate the dashboard efficiently.

Service Dashboard

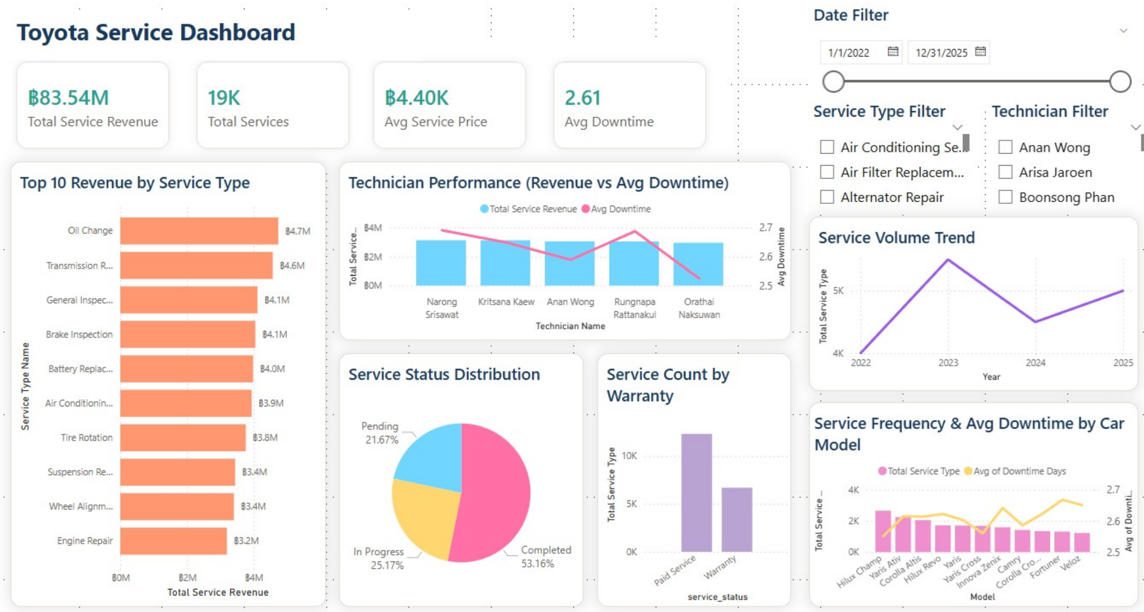


Figure 29: Toyota Service Dashboard

The Service Dashboard is designed to monitor operational performance in after-sales services as shown in Figure 29. The layout begins with key KPIs, including Total Service Revenue, Total Services, Average Service Price, and Average Downtime, placed at the top to provide a quick summary of service performance.

The dashboard uses a variety of visualization types to represent different aspects of service data. **Bar charts** are used to compare revenue by service type and technician performance, allowing clear comparison across categories. A **line chart** is used to illustrate service trends over time, helping users track changes in service volume. A **pie chart** is used to display service status distribution, making it easy to understand the proportion of completed, pending, and in-progress services.

Interactive **filters**, including date, service type, and technician, allow users to refine the analysis based on specific conditions. The layout is organized into sections, grouping related charts together, and uses consistent formatting and color schemes to ensure clarity and usability.

Inventory Dashboard

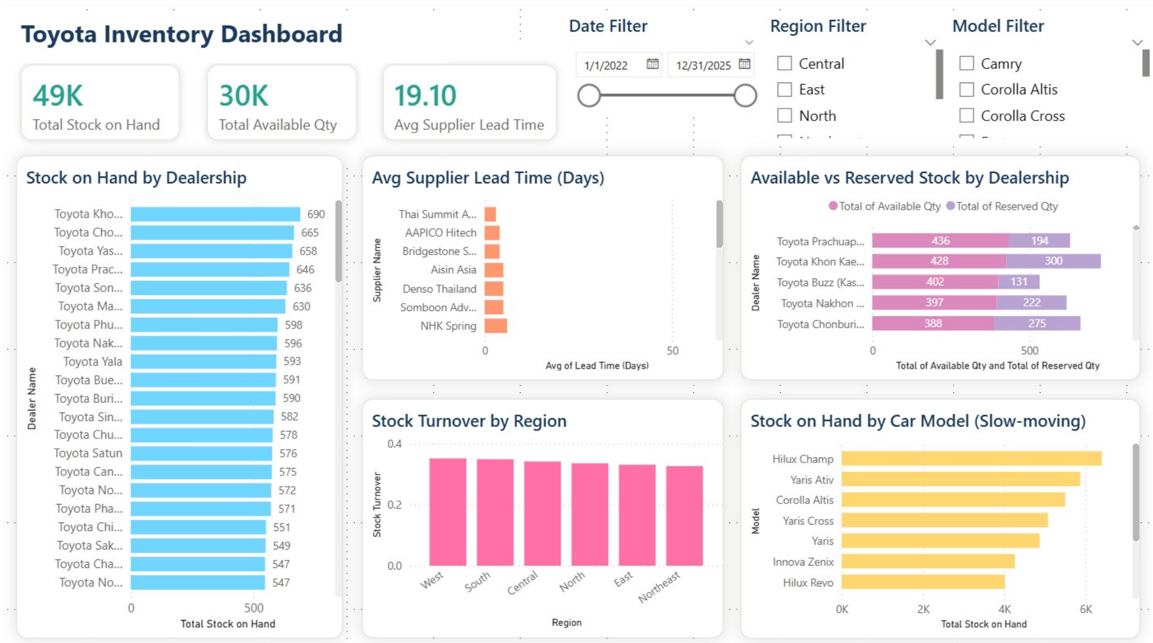


Figure 30: Toyota Inventory Dashboard

The Inventory Dashboard in Figure 30 is structured to provide insights into stock management and supply chain performance. Key KPIs, such as Total Stock on Hand, Total Available Quantity, and Average Supplier Lead Time, are displayed at the top for immediate visibility.

The dashboard primarily uses **bar charts** to compare stock levels across dealerships and car models, as well as to highlight slow-moving inventory. Additional comparison charts are used to display available versus reserved stock, supporting easy side-by-side analysis. A **column chart** is used to show stock turnover by region, allowing users to compare performance across geographic areas.

Interactive **filters**, including date, region, and model, are incorporated to enhance user interaction and allow flexible data exploration. The design maintains a consistent color scheme and clear labeling throughout, ensuring readability and a professional appearance. The layout is clean and structured, enabling users to quickly locate and interpret key information.

6. Discussion

Benefits

- **Integrated Business Analysis:** Consolidating data from sales, service, and inventory into a unified star schema enables cross-functional analysis across all three business domains within a single platform, eliminating the need to analyze each process separately.
- **Actionable KPI Monitoring:** Interactive dashboards provide continuous visibility into key metrics such as Total Revenue, Average Downtime, and Stock Turnover, allowing managers to monitor performance and respond to changes promptly.
- **Enhanced Decision Support:** Power BI's filtering and drill-down capabilities allow users to explore data from company-wide overview down to individual dealership, model, or technician level, supporting both strategic and operational decisions.
- **Scalability and Flexibility:** The star schema design follows industry-standard principles, making it straightforward to extend with additional fact tables or dimensions in the future without restructuring the existing model.

Challenges

- **Data Quality and ETL Complexity:** Since datasets were generated independently by multiple team members, inconsistencies in data types and text formatting were encountered and required correction in Power Query Editor before analysis could proceed reliably.
- **Synthetic Data Limitations:** Artificially generated data produced overly uniform results in some visualizations, such as nearly identical stock turnover ratios across all regions, which limits the depth of insight that can be drawn from certain analyses.
- **Fact Table Design Adjustment:** Determining the appropriate grain for each fact table required careful consideration to avoid double-counting and ensure accurate aggregation, particularly for service duration calculations in Fact_Service.

Limitations

- **No Real Operational Data:** All datasets are synthetic, meaning insights and recommendations are illustrative rather than reflective of actual Toyota business performance.
- **No Real-Time Processing:** The solution relies on static CSV files and does not support live data refresh. A production environment would require scheduled or real-time database connectivity.
- **Incomplete Dashboard Coverage:** Certain analytical areas such as customer repurchase behavior and salesperson performance trends were not fully explored due to time and data constraints, representing opportunities for future development.

7. Conclusion

This project successfully developed a Business Intelligence solution for Toyota's dealership operations by designing a dimensional data warehouse and implementing interactive Power BI dashboards. The star schema model, comprising three fact tables — Fact_Sales, Fact_Service, and Fact_Inventory — effectively consolidated data across sales, after-sales service, and inventory management domains.

Key findings revealed that revenue fluctuated across the analysis period, with a notable decline in 2024 followed by a strong recovery in 2025. Sales performance is concentrated among a few top-performing car models and dealerships, with Hilux Champ generating the highest revenue and the Central region leading in overall sales contribution, indicating an uneven distribution of revenue that presents both risk and opportunity. Service operations emerged as a stable and recurring revenue stream, with Oil Change and Transmission Repair being the highest-revenue service types, and non-warranty paid services accounting for the majority of service transactions. Inventory analysis exposed imbalances in stock distribution across dealership locations, with certain models such as Hilux Champ holding disproportionately high stock levels relative to others, highlighting the need for more strategic inventory allocation and supplier management.

From this project, the team gained practical experience in applying data warehousing concepts such as grain definition, dimensional modeling, and ETL processes. The use of Power BI reinforced skills in data visualization, KPI design, and dashboard interactivity.

For future improvements, the system could be enhanced by integrating real operational data to improve analytical accuracy. Implementing real-time data processing pipelines would enable more timely decision-making. Additionally, expanding the dashboard scope to include customer loyalty analysis, financial forecasting, and predictive analytics would further increase the strategic value of the BI solution. Overall, this project demonstrated that even with synthetic data, a well-designed data warehouse and BI solution can provide meaningful insights and serve as a scalable foundation for data-driven decision-making in the automotive industry.

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